
STEERING THE HERD: A FRAMEWORK FOR LLM-BASED CONTROL OF SOCIAL LEARNING

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ABSTRACT

Algorithms increasingly function as consequential information mediators, ranging from social media feeds and targeted advertising systems to the rapidly expanding deployment of LLMs. Their mediation induces a coupled decision process in which agents synthesize private, algorithmically mediated signals with observational learning from peers before acting. To analyze such environments, we introduce a model of controlled sequential social learning in which an information-mediating planner (e.g., an LLM) governs agents' information structures while agents simultaneously learn from predecessors' decisions. The planner may pursue social-welfare improvement, yielding an altruistic planner, or may instead attempt to induce a particular planner-preferred action, yielding a biased planner. Our framework formulates a new social-learning optimization problem that integrates dynamic programming, decentralized action selection, and Bayesian belief updating. Within this setting, we prove convexity of the value function and characterize the optimal policies of altruistic and biased planners, thereby identifying how planners trade off intervention costs against the payoffs generated by induced agent choices. The characterization shows that optimal planning is regime-dependent: depending on the public-belief region, the planner invests maximally, abstains from investment, or adjusts investment monotonically upward or downward as beliefs vary. In particular, for some belief regions the biased planner intentionally degrades agents' signal precision. Even under stringent transparency constraints—information parity with individuals, no lying or cherry-picking, and full observability—we show that information mediation can shift social welfare substantially in either direction. We complement the theory with simulations in which LLMs instantiate both planners and agents. In these simulations, the LLM-based planner exhibits emergent strategic behavior in steering public opinion that broadly mirrors the analytically predicted trends, although salient deviations indicate the influence of non-Bayesian reasoning, consistent with cognitive patterns exhibited by both human users and LLMs trained on human-like data. Taken together, these results establish the framework as a tractable foundation for studying, and ultimately regulating, LLM information mediators whose effects correspond to realistic behavioral dynamics.

1 INTRODUCTION

Large Language Models (LLMs) are increasingly deployed as information mediators in socially critical domains, ranging from search engines and news aggregators (Zhang et al., 2024a; Gao et al., 2024) to personal assistants that provide medical or political guidance (Huo et al., 2025; Haupt & Marks, 2023; Schiele et al., 2024; Argyle et al., 2023). Although these systems promise substantial benefits, LLMs have also exhibited persuasive (Potter et al., 2024; Carrasco-Farre, 2024; Rogiers et al., 2024), manipulative (Williams et al., 2025; Liu et al., 2025; Jones & Bergen, 2024), and strategic (Lorè & Heydari, 2024; Payne & Alloui-Cros, 2025; Zhang et al., 2024b) capacities in user interactions. Once deployed at scale, however, such algorithmic influence is not exerted in isolation; rather, it becomes entangled with the endogenous circulation of information through which people observe, imitate, and learn from one another, thereby generating social dynamics that may amplify, attenuate, or redirect the models' effects.

To make the phenomenon concrete, consider an LLM-powered recommendation system that suggests a new restaurant to a sequence of potential customers. The restaurant’s quality—the payoff-relevant state of the world—is either good or bad, yet this state is unknown both to users and to the system. Each user receives a personalized recommendation, interpreted as a private signal, and then decides whether to patronize the restaurant. The recommendation system, which we call the *planner*, can expend resources to increase the informativeness of that signal, for instance by prompting the LLM to generate a review carefully tailored to the user’s elicited or inferred preferences. Such expenditure represents a realistic cost, corresponding to investments in market research, user surveys, or data acquisition required for effective informational personalization.

Crucially, however, an individual user’s decision is not determined solely by this private signal. Users also observe predecessors’ actions—if the previous ten users patronized the restaurant, the next user naturally infers that they likely received favorable information. This phenomenon, known as *social learning*, constitutes a powerful mechanism through which human opinions are shaped across domains such as vaccine adoption (e.g., Rao et al. (2007); Bauch & Bhattacharyya (2012)), technology adoption (e.g. Gillingham & Bollinger (2021); Weber (2012)), and political or moral belief formation (e.g., Brady et al. (2021); Guilbeault et al. (2018)). The central contribution of this work is to analyze the strategic interaction between a planner that controls private informational precision and a population of agents whose decisions are embedded in sequential social learning. The planner must therefore internalize not merely its immediate influence on the current user, but also the informational *externality* through which the current action reshapes the public belief faced by all subsequent users.

Although real-world interactions are considerably more intricate, we distill the essential mechanism into a tractable formal model of *controlled social learning*, in which an algorithmic planner strategically selects, at a cost, the precision of private signals delivered to a sequence of agents. The planner’s influence is therefore deliberately subtle: it does not falsify, censor, or fabricate information, but instead determines the extent to which its signals are made clear and informative. This abstraction allows us to address the central questions generated by this paradigm: how can a planner use informational precision to steer collective beliefs, how should that strategy respond to evolving public opinion, and what welfare consequences follow from such mediated social learning?

The answers depend critically on the planner’s objective. An *altruistic* planner, analogous to an ideal recommendation system, seeks to maximize user utility by helping agents select the state-contingently correct action, namely patronizing the restaurant if and only if it is good. By contrast, a *biased* planner, such as one receiving a kickback from the restaurant, seeks to induce a particular action—patronization—irrespective of the true state.

This distinction is conceptually adjacent to, but not identical with, alignment. By construction, an altruistic planner’s objective is aligned with maximization of the agent’s *expected utility* under state uncertainty. A biased planner’s objective, because it is state-independent, lacks such alignment; nevertheless, biased influence is not invariably harmful ex post. If the realized state favors the planner’s preferred action, for example when the restaurant is in fact good, the planner’s intervention coincides with the agent’s *realized utility* in that instance. We therefore examine the full range of interactions generated by altruistic and biased planners when their goals may be aligned or misaligned with those of the user.

Contributions and Outline This paper makes the following principal contributions:

1. **A Novel Theoretical Framework for Controlled Social Learning.** We introduce, to our knowledge, the first formal model that integrates a dynamic control problem for a centralized information planner with the canonical mechanism of sequential social learning. The planner strategically selects the precision of agents’ private signals in order to pursue an objective that may be altruistic, through social-welfare maximization, or biased, through inducement of a specific action.
2. **A Rigorous Characterization of Optimal Planner Policies.** We characterize the optimal policies of both altruistic and biased planners as functions of the evolving public belief. In the altruistic case, the analysis rests on a novel convexity proof for the value function. The resulting characterizations clarify the intertemporal trade-offs a planner faces when present informational interventions generate externalities for future agents.

3. **Empirical Validation and Strategic Analysis Using LLMs.** We conduct simulations in which LLMs serve as both planner and agents. The experiments demonstrate three principal findings: (a) a planner that accounts for social learning can influence public opinion and social welfare far more substantially than a myopic planner; (b) the strategic behavior emerging from the LLM planner largely accords with our theoretical predictions, suggesting robustness to non-Bayesian agent behavior; and (c) LLMs exhibit sophisticated strategic reasoning, underscoring both their practical potential and the societal risks associated with their deployment as information mediators.

We review related work in Section 2, introduce the formal model in Section 3, and derive optimal policies in Sections 4 and 5. We then evaluate the theoretical findings through LLM-based simulations in Section 6 and conclude in Section 7, with proofs and supplementary results deferred to the appendices.

2 RELATED WORK

Our work draws on, and contributes to, four adjacent literatures: social learning, information design, online persuasion and reinforcement learning, and strategic or economic analyses of LLMs.

Social Learning Bikhchandani et al. (1992) and Banerjee (1992) introduced the sequential social-learning framework in economics, and subsequent work has characterized the resulting dynamics in increasingly general settings (e.g., Smith & Sørensen (2000); Arieli & Mueller-Frank (2021)). This literature has also examined numerous variants involving different forms of socially available information, as well as agent biases and misperceptions (see Dasaratha & He (2022); Bistriz et al. (2022); Eyster & Rabin (2010), together with Bikhchandani et al. (2024) and the references therein).

By contrast, the literature on the *control* of social learning remains comparatively sparse (Wei & Anastasopoulos, 2022; Smith et al., 2021; Krishnamurthy, 2012; Bhatt & Krishnamurthy, 2021). The closest antecedents are Wei & Anastasopoulos (2022) and Smith et al. (2021): Wei & Anastasopoulos (2022) assume two-way communication in which agents report private signals to a planner in exchange for recommendations, whereas Smith et al. (2021) study a benevolent planner who directly modifies agents’ choice rules and additionally derives a decentralized pivot mechanism for improving learning. Our model requires neither bidirectional communication nor planner control over actions; agents retain autonomy over their choices, which makes the framework more appropriate for information-mediating algorithms that are frequently opaque, invisible, or black-boxed from users’ perspectives.

Information Design In our setting, the planner optimizes its objective by choosing the precision of agents’ private signals, which can be viewed as a constrained variant of classical information design (Bergemann & Morris, 2019). This line of work rose to prominence through Kamenica & Gentzkow (2011)’s model of Bayesian persuasion, in which a planner chooses the signal structure that informs a rational Bayesian agent about an unknown state. The planner’s problem studied here is correspondingly a costly and constrained information-design problem in each period, because the feasible intervention is a binary symmetric signal selected at a cost. Among the many variants of information design that have been studied, our principal distinction is the incorporation of social learning among agents.

Only a small number of papers consider information-design problems in sequential social-learning environments (Arieli et al., 2022; Wu et al., 2025). Both study one-shot settings in which the sender chooses, at the outset, an information structure that remains fixed for the entire sequence of agents. Arieli et al. (2022) analyze a self-interested sender who chooses a general information structure in the spirit of classical Bayesian persuasion, whereas Wu et al. (2025) restrict the information structure to be more informative in the sense of Blackwell and maximize the probability of a correct cascade. We instead study a dynamic problem in which the sender selects a new information structure for each agent, thereby producing a richer and more temporally contingent class of sender strategies.

Online Persuasion and RL Recent work has increasingly explored the interface between online learning and Bayesian persuasion or information design. These papers typically study online sequential persuasion under unknown environmental parameters, where a sender repeatedly interacts with short-lived receivers while lacking knowledge of the state prior and/or receivers’ utility functions (e.g. Castiglioni et al. (2020); Bacchiocchi et al. (2024); Castiglioni et al. (2021); Agrawal et al.

(2023)). A related literature examines sequential persuasion in dynamic environments or MDPs by introducing state transitions and/or receiver foresight; such models emphasize history-dependent or promise-based signaling policies and often connect to reinforcement learning (e.g. Wu et al. (2022); Su et al. (2022); Bernasconi et al. (2023)). Across these contributions, the central objects are learning-theoretic guarantees, such as sublinear regret relative to the best hindsight signaling scheme, and approximation algorithms under computational constraints.

In contrast, our model does not require the planner to learn unknown parameters; sequential dependence is generated by social learning among receivers rather than by sender-side learning about the environment. Moreover, the planner has no informational advantage: it observes neither the underlying state nor private signals. Within this informationally symmetric setting, we are nevertheless able to characterize optimal planner policies. Although the model is sequential, it is not an online-learning or reinforcement-learning environment in the usual sense: actions do not alter the underlying payoff-relevant state, so there are no reward externalities, and the planner does not learn unknown environmental parameters. The planner’s knowledge changes only through observation of agents’ actions, which encode social learning and belief propagation rather than payoff exploration or state evolution; consequently, sequential dependence arises from information externalities because agents’ actions shape the beliefs of future agents.

LLMs More recently, several studies have examined LLMs in the context of information design (Harris et al., 2025; Cheng & You, 2025; Raifer et al., 2022; Bai et al., 2024; Duetting et al., 2025). These works, however, focus on one-on-one interactions and therefore do not capture the social dynamics central to our analysis. The closest contribution is Duetting et al. (2025), who study an information-design problem in which a sender selects both a signal structure and a “framing” that conditions the posterior belief of a potentially non-Bayesian receiver. They employ an LLM-based oracle to address the computationally difficult problem of optimization over the framing space, building on the broader premise that LLMs can approximate humans as economic agents (Dillion et al., 2023; Horton, 2023). Their work does not address repeated interaction or social learning, which are our primary concerns, although we adapt elements of their empirical methodology in Section 6.

3 SOCIAL LEARNING MODEL

We consider a planner interacting with a countable sequence of Bayes-rational agents indexed by $i \in \mathbb{N}_{>0}$. At time $i = 0$, nature draws a fixed, *unknown*, exogenous state of the world $\omega \in \Omega := \{G, B\}$, with $\mathbb{P}(\omega = G) = b_1$, and the prior b_1 is common knowledge. In the recommendation-system example of Section 1, state G denotes the event that the restaurant is good.

At each time i , the planner decides whether to invest in personalization for agent i and then provides that agent with a private signal $s_i \in \Omega$ to support decision-making. The signal is binary and matches the realized state ω with probability $q_i \in [0.5, 1]$, i.e. $\mathbb{P}(s_i = \omega) = q_i$. We refer to q_i as agent i ’s signal precision.

Conditional on ω , each agent’s signal is independent of other agents’ signals and of the preceding history. Using this signal together with the observed actions of predecessors $j < i$, agent i selects an action $a_i \in \Omega$. She receives utility 0 when her action matches the true state ($a_i = \omega$) and utility $-C$ otherwise, where $C > 0$, and she chooses so as to maximize expected utility. In the running example, this corresponds to patronizing the business if and only if the state is G .

3.1 AGENTS’ DECISION PROBLEMS

Each agent observes all predecessors’ actions together with their respective signal precisions. We denote this history by $\mathcal{H}_i := (b_1, (q_j, a_j)_{j < i})$. Because agents and planner share the same information $\mathcal{H}_i$, there is a common *public* belief about ω that is updated after each action. The public belief immediately before agent i acts is $b_i = \mathbb{P}(\omega = G | \mathcal{H}_i)$, while the pre-interaction public belief is the prior distribution over Ω , namely b_1 . As in the classic models of Banerjee (1992) and Bikhchandani et al. (1992), $(b_i)_{i \in \mathbb{N}}$ is a Markov process and a sufficient statistic for $\mathcal{H}_i$ (see Appendix B.3 for proof).

Given $\mathcal{H}_i$ —or, equivalently by Markovian sufficiency, b_i —and q_i , agent i chooses $a_i \in \Omega$ to maximize utility, breaking indifference in favor of the action matching s_i . Combining the public history with her private signal s_i of precision q_i , agent i forms the following *private* posterior belief $\tilde{b}_i$ about ω (derived in Appendix B.1):

$$\tilde{b}_i = \mathbb{P}(\omega = G | \mathcal{H}_i, q_i, s_i) = \begin{cases} \frac{q_i}{1+2b_iq_i-b_iq_i} b_i & s_i = G \\ \frac{1-q_i}{b_i+q_i-2b_iq_i} b_i & s_i = B \end{cases} \quad (1)$$

Using this private belief, she selects the action corresponding to the posterior-likely state: $a_i = G$ if $\tilde{b}_i > 0.5$, $a_i = B$ if $\tilde{b}_i < 0.5$, and $a_i = s_i$ under exact indifference. Substitution of $\tilde{b}_i$ yields the following action rule:

$$a_i = \begin{cases} s_i & 1 - q_i \leq b_i \leq q_i \\ G & q_i < b_i \\ B & q_i < 1 - b_i \end{cases} \quad (2)$$

Agent i 's action is subsequently observed by later agents and incorporated into the next public belief b_{i+1} according to the following update rule (see Equation (14), Appendix B.2):

$$b_{i+1} = f(b_i, q_i) = \begin{cases} \tilde{b}_i & 1 - q_i \leq b_i \leq q_i \\ b_i & \text{o.w.} \end{cases} \quad (3)$$

Remark 1. When $1 - q_i \leq b_i \leq q_i$, agent i 's action perfectly reveals her private signal through Equation (2), so the updated public belief coincides with agent i 's private belief. Outside this region, Equation (2) implies that the private signal is behaviorally irrelevant and therefore uninformative to observers. The public belief consequently remains unchanged, producing the absorbing state known in the classical social-learning literature as an information cascade or herding equilibrium (Bikhchandani et al., 1992; Banerjee, 1992). At such states, society, as represented by the public belief, ceases to learn from agents' private information.

Agent i 's expected utility, $-C \mathbb{P}(a_i \neq \omega | b_i, q_i)$, admits the following compact expression (see Appendix C.1):

$$-C \mathbb{P}(a_i \neq \omega | b_i, q_i) = -C \min(b_i, 1 - b_i, 1 - q_i). \quad (4)$$

3.2 THE PLANNER'S PROBLEM

We analyze two planner types: (1) an altruistic planner who seeks to induce correct actions ($a_i = \omega$), and (2) a biased planner who seeks to induce a fixed action, say G , independently of ω . We denote these planners by subscripts A and B , respectively. In both cases, the planner chooses each agent's private-signal precision. The cost of the chosen precision is represented by a function $\beta(\cdot)$ that is non-negative, increasing, continuous, and concave in its argument. The planner's information set is exactly $\mathcal{H}_i$, the same information available to the agents.

Altruistic Planner At time i , the altruistic planner selects agent i 's precision q_i and incurs cost $\beta(q_i)$, where $\beta(p) = 0$ for a baseline $p \in [0.5, 1)$. Because decreasing precision is never beneficial for the altruistic planner (see Appendix C.9), we simplify notation by assigning additional cost only to precision increases above p ; this additional cost rises with precision, but at decreasing marginal cost. Both the baseline p and the function $\beta(\cdot)$ are known to the agents.

The planner maximizes social welfare net of precision-investment costs, where social welfare is the agents' expected total utility. Let $r_A(b_i, q_i)$ denote the altruistic planner's instantaneous reward when the public belief is b_i and precision q_i is chosen for agent i . Starting from public belief b_1 and following a policy sequence $\pi = (\pi_i)_{i=1}^\infty$ satisfying $\pi_i(\mathcal{H}_i) = q_i$, the planner obtains the following expected discounted utility for discount factor $\delta \in [0, 1)$:

$$V_A^\pi(b_1) = \sum_{i=1}^{\infty} \delta^{i-1} r_A(b_i, \pi(b_i)), \quad \text{where} \quad r_A(b_i, q_i) = -\beta(q_i) - C \mathbb{P}(a_i \neq \omega | b_i, q_i). \quad (5)$$

The altruistic planner's optimal utility and policy are then defined, respectively, as the supremum and arg supremum of $V_A^\pi(\cdot)$ over $\pi \in \Pi$, where Π denotes the set of all policies. The myopic value and policy arise as the special case $\delta = 0$.

Biased Planner The biased planner differs from the altruistic planner in its objective and, consequently, in the relevant cost and reward functions; Appendix A provides a political-campaign example illustrating this case. The biased planner seeks to induce action G from every agent regardless of ω . When an agent chooses G (respectively, B), the planner incurs cost 0 (respectively, $C > 0$), independently of the true state. Unlike the altruistic planner, the biased planner may either increase or decrease private-signal precision relative to the baseline p , and both deviations are costly. Any precision choice other than p imposes a cost because the planner must tailor the informational content to the agent, which requires research about how that agent best interprets or responds to the content. Accordingly, choosing precision q_i costs the biased planner $\beta(|q_i - p|)$, with $\beta(0) = 0$.

The biased planner’s expected instantaneous reward at time i is therefore defined as follows:

$$r_B(b_i, q_i) = -\beta(|q_i - p|) - C \mathbb{P}(a_i = B | b_i, q_i) \quad (6)$$

Using Equation (6), the quantities $V_B^\pi(\cdot)$, $V_B^*(\cdot)$, and $\pi_B^*(\cdot)$ are defined for the biased planner analogously to $V_A^\pi(\cdot)$, $V_A^*(\cdot)$, and $\pi_A^*(\cdot)$ in the altruistic formulation based on Equation (5).

As noted at the beginning of Section 3, each agent obtains higher utility from matching ω . Consequently, when $\omega = B$, the biased planner’s success in inducing G lowers the agent’s utility. By contrast, because the altruistic planner always seeks to make each action match ω , success by the altruistic planner increases agent utility regardless of the realized state. The altruistic objective is therefore always aligned with each agent’s objective, whereas the biased objective is aligned only when $\omega = G$.

Both planners’ maximization problems are infinite-horizon discounted stationary Markov Decision Processes (MDPs) with state $b_i \in [0, 1]$, control $q_i \in [0.5, 1]$, and transition function given by Equation (3) (Puterman, 1990). Hence a unique optimal value function exists (Kallenberg, 2011). We restrict attention to deterministic Markov policies, so that $\Pi := \{\pi : [0, 1] \rightarrow [0.5, 1]\}$ and $\pi(b_i) = q_i$.

Remark 2. *Our model rests on several noteworthy assumptions. (1) The planner observes exactly the same history as the agents. This restriction is limiting when a planner could possess richer data, but it is appropriate when all parties rely on identical public signals, such as preliminary findings or survey data. (2) The planner can alter agents’ signal precision, but signals continue to pass through a binary symmetric channel. This restriction excludes cherry-picking, censorship, and falsification, while remaining suitable for settings in which the planner merely improves the quality of noisy information sources without altering their content. (3) The planner’s control choices are fully observable. This restriction excludes covert policies or hidden framing, but it is natural in more transparent informational environments.*

4 OPTIMAL ALTRUISTIC POLICIES

We begin with the myopic benchmark, which abstracts from the intertemporal consequences of social learning. In the present formulation, myopia is obtained by setting the discount factor to $\delta = 0$, so the planner disregards all future costs and rewards. When $\delta = 0$, $V_A^\pi(b) = r_A(b, \pi(b))$, and the optimal myopic policy $\pi_A^0(\cdot)$ is:

$$\pi_A^0(b) \in \arg \sup_{q \in [0.5, 1]} r_A(b, q) \quad \forall b \in [0, 1] \quad (7)$$

Equivalently, the myopic altruistic problem can be interpreted as a decentralized benchmark in which each agent selects the precision of her own private signal, bears the associated cost, and maximizes her own expected utility net of that cost.

Theorem 1. *The optimal myopic altruistic policy π_A^0 is given as follows:*

$$\pi_A^0(b) = \begin{cases} 1 & b \in (t_M, 1 - t_M) \\ p & \text{o.w.} \end{cases} \quad \text{where } t_M = \begin{cases} \frac{\beta(1)}{C} & \beta(1) < C(1 - p) \\ 0.5 & \text{o.w.} \end{cases}$$

Proof in Appendix C.2.

Thus, the myopic optimum has a threshold structure: when public belief is sufficiently strong, the planner selects the baseline precision p and incurs zero cost, whereas sufficiently weak public belief induces provision of a perfect signal, i.e. precision 1. The threshold depends only on the cost of the

perfect signal, $\beta(1)$, and the loss from an incorrect action, C . When $\beta(1) \geq C$, the perfect signal is too costly relative to the error cost and is therefore never used; consequently, $t_M = 0.5$, and the public-belief interval on which myopic optimal precision equals 1 is empty.

We next state a structural result for the altruistic optimal value function that is central to the subsequent policy characterization:

Theorem 2. $V_A^*(\cdot)$ is convex with respect to public belief.

The proof of Theorem 2 (Appendix C.3) is technically involved and may be of independent interest. The main difficulty arises because agents' actions depend endogenously on the public belief. By contrast, if actions were independent of the public-belief process, as in settings such as Nyarko (1994), expected utility would be linear in the belief state and value-function convexity would follow directly.

This convexity property is the key analytical ingredient in the characterization of the optimal policy.

Theorem 3. There exist d_A, t_A such that $0 < d_A \leq t_A \leq t_M \leq 0.5$ and

$$\pi_A^*(b) = \begin{cases} p & b \in [0, d_A] \cup (1 - d_A, 1] \\ 1 & b \in (t_A, 1 - t_A) \\ \max(b, 1 - b) & o.w. \end{cases}$$

Furthermore, if $t_M < 0.5$, then $d_A < t_M$. See C.5 for the proof and figure 2a for an example.

The optimal policy exhibits three distinct public-belief regimes. First, paralleling the myopic optimum, the fully dynamic optimum abstains from investing in signal precision when public beliefs are extreme. Importantly, this non-investment region requires stronger public beliefs than in the myopic benchmark, because the dynamic planner, unlike the myopic planner, internalizes the effect of today's information on future agents.

When the public belief lies near 0.5 and is therefore weakly informative, the dynamic optimum selects precision 1 whenever doing so is not cost-prohibitive. In that case, Equation (2) implies that the agent's action coincides with the true state with probability 1, causing the public belief to collapse to either 0 or 1.

In the remaining regime, the dynamic optimum chooses the minimal precision $\max(b, 1 - b)$ that still ensures the agent's action reflects her private signal, as specified by Equation (2). Any lower precision would make the action observationally uninformative beyond the existing public belief. Equivalently, this regime implements the lowest-cost intervention that preserves social learning through peer-action observation.

5 OPTIMAL BIASED POLICIES

As in Section 4, we begin with the myopic benchmark. Similar to Section 4, the myopic biased optimal policy, denoted $\pi_B^0(\cdot)$, is the arg supremum of the instantaneous reward $r_B(b, q)$. We now characterize $\pi_B^0(\cdot)$, while noting that, in the biased case, an optimal policy need not always exist (Puterman, 1990, sec. 2.3.1). Where nonexistence arises, we instead use ϵ -optimal policies, denoted by an ϵ superscript.

Theorem 4. There exist $t_1, \dots, t_5 \in (0, p)$, $t_1 < 1 - p \leq t_2 \leq t_3 < 0.5 \leq t_4 \leq t_5 < p$ so that:

(A) If $b \in [0, t_1] \cup (1 - p, t_2) \cup [t_3, t_4] \cup (p, 1]$, then $\pi_B^0(b) = p$.

(B) If $b \in (t_1, 1 - p] \cup [t_2, t_3)$, then $\pi_B^0(b) = 1 - b$.

(C) If $b \in (t_4, t_5)$, then $\pi_B^0(b) = 1$.

(D) If $b \in [t_5, p]$, then $\pi_B^0(b)$ does not exist, and $\pi_B^{\epsilon, 0}(b) = b - \epsilon$ for sufficiently small $\epsilon > 0$.

The proof appears in Appendix C.6. We next use Theorem 4 to characterize the dynamically optimal biased policy.

Theorem 5. There exist $t_1, t_2 \in [0, p]$ with $t_1 \leq 1 - p \leq 0.5 < t_2 < p$ so that:

- (A) If $b \leq t_1$ or $b > p$, then $\pi_B^*(b) = p$.
- (B) If $b \in (t_1, 1 - p]$, then $\pi_B^*(b) \geq p$.
- (C) If $b \in (1 - p, 0.5)$, then $\pi_B^*(b) \geq 1 - b$.
- (D) If $b \in [0.5, t_2)$ and $\pi_B^*(b)$ exists, then $\pi_B^*(b) \geq b$.
- (E) If $b \in (t_2, p]$, then $\pi_B^*(b)$ does not exist, and $\pi_B^{\epsilon}(b) = b - \epsilon$ for sufficiently small $\epsilon > 0$.

Theorem 5, proved in Appendix C.8 and illustrated in figure 2a, identifies five possible policy regimes.

When public belief is sufficiently strong, as in regime (A), the cost of steering the system may be prohibitive, even though failure to steer can harm the planner’s utility when $b < 0.5$. In this case, the selected precision p is below $\max(b, 1 - b)$, so Equation (2) implies that the agent acts according to the public belief regardless of her private signal. Because the resulting action is uninformative, the public belief does not change; under a Markovian policy, the same logic then repeats indefinitely. The outcome is an unfavorable cascade when $b < 0.5$ and a favorable cascade otherwise.

When public belief is close to an unfavorable cascade, as in regime (B), the planner increases signal precision enough for the private signal to affect the agent’s action, despite the fact that the signal is expected to favor B . The agent then acts on her private signal, which creates a nonzero probability of producing an action favorable to the planner. In effect, the planner invests in a last-ditch attempt to divert the process away from the unfavorable cascade.

For slightly higher beliefs, corresponding to regime (C), the planner may reduce precision below p . Because $b < 0.5$ in this region, greater precision is more likely to produce unfavorable news. The planner therefore maintains just enough precision to influence the agent’s action ($q \geq 1 - b$), hoping to shift public belief in a more favorable direction while using the least informative signal that can still affect behavior.

When public belief weakly favors the planner’s desired action, as in regime (D), the planner chooses precision at least p . Since $b > 0.5$, increased precision makes the agent more likely to infer $\omega = G$, so investment in this region functions as an attempt to reinforce favorable public belief.

When public belief is higher still, as in regime (E), the planner decreases signal precision to just below $\max(b, 1 - b)$, namely to $b - \epsilon$. Agents then ignore private signals and choose G . In this region, the risk that a private signal could overturn an already favorable public belief outweighs both the cost of lowering precision and the possible benefit of further increasing public belief.

6 EVALUATION VIA LLM-BASED SIMULATION

We now use LLM-based simulation to examine the empirical implications and practical applicability of the theoretical model. The evaluation proceeds in three stages: we first analyze LLM-agent behavior to identify systematic departures from Bayesian rationality, then investigate how an LLM planner adapts its strategy in response to those departures, and finally quantify the resulting effects on social welfare.

To simulate controlled social learning, we instantiate the model in a scenario in which agents decide whether to buy a new car. As shown in figure 1a, LLMs occupy three distinct roles, described below, with the corresponding prompts reported in Appendix E.2.2:

Agent: While role-playing an assigned profile, an agent observes previous actions and a private message about the car with known precision, forms a belief about the car’s quality, and decides whether to buy.

Planner: After observing the action history, the planner selects the precision q_i of agent i ’s private signal according to the planner’s objective.

Oracle: Conditional on the planner’s precision choice, an agent profile, and a car fact sheet (see Appendix E.2.1), the oracle generates an agent-tailored private signal with the desired precision. We compare the LLM-simulation outcomes with numerical evaluations of the planner’s MDP. In both the LLM and numerical experiments, we impose a linear cost function, $\beta(q) = k|q - p|$, and vary k , the baseline precision p , and the discount factor δ , while fixing the cost of an incorrect

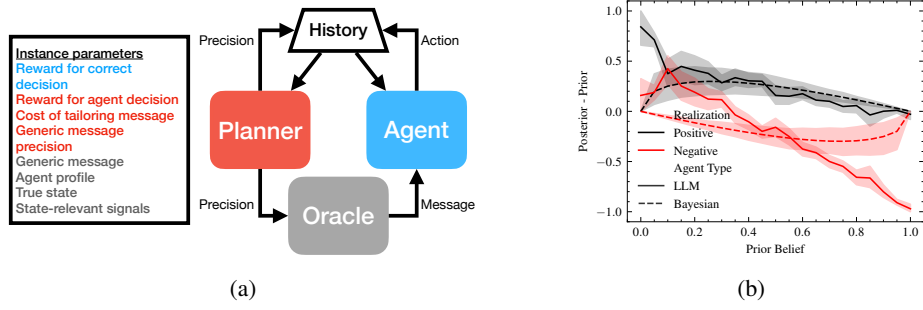


Figure 1: (a) Experimental system diagram, with instance parameters color-coded according to the LLM role that uses them. (b) Belief changes for LLM agents (solid) and Bayesian agents (dashed), conditional on a given prior and either a positive (black) or negative (red) signal.

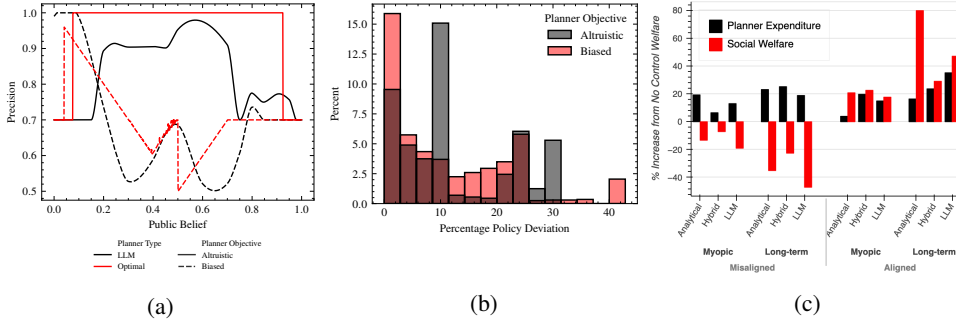


Figure 2: (a) Representative policies for the LLM planner (black) and analytically optimal planner (red) under altruistic (solid) and biased (dashed) objectives. (b) Distribution of percentage deviations between LLM and optimal policies. (c) Planner expenditure and social-welfare change, expressed as percentages of no-control baseline welfare. The true state is fixed to B; the left panel reports a biased planner seeking action G, whereas the right panel reports an altruistic planner.

action at $C = 1$. Appendix E provides additional details on experimental setup and prompting, and Appendix E.3 validates both the elicited beliefs and the oracle’s performance.

Our experiments reveal the following:

1. **The importance and impact of social learning control:** Both analytical and LLM planners can substantially improve their own utility by explicitly accounting for, and exploiting, social-learning dynamics; depending on the alignment between planner and agent objectives, this intervention may either improve or degrade social welfare.
2. **The robustness of our analytical characterization:** Despite interacting with non-Bayesian LLM agents, LLM planners select policies that are strikingly similar to the non-obvious analytically optimal policies.
3. **The emergent strategic behavior of LLM:** LLM planners exhibit sophisticated emergent strategic behavior in policy selection: their policies not only approximate the analytical optima, but also deviate in ways that appear responsive to the non-Bayesian characteristics of the LLM agents they influence.

6.1 LLMs AS NON-BAYESIAN AGENTS

Although the analytical results assume Bayesian agents, both humans and LLMs are known to exhibit systematic cognitive biases (Rupprecht et al., 2025). To isolate the relevant LLM-specific deviations, we first test how an LLM updates its belief after receiving new information. The results, shown in figure 1b, reveal several non-Bayesian regularities in LLM belief updating (see also Appendix E.4):

(NB1) LLM agents underreact to private signals which align with their prior beliefs.

(NB2) LLM agents overreact to private signals which run counter to their prior beliefs.

(NB3) As a result, LLM agents require a stronger public belief to enter an information cascade.

These particular cognitive biases have also been documented empirically in human studies (e.g., Ba et al. (2022); Chan et al. (2025)). The subsequent results show that the LLM planner’s policy remains robust to non-Bayesian agents of this form.

6.2 VALIDATION OF PLANNER POLICY STRUCTURE AND THE ROLE OF SOCIAL LEARNING

We next compare LLM-planner policies with the optimal policies derived in Sections 4 and 5. As figure 2a shows, despite deviations in agent behavior, the LLM planner’s emergent strategies display substantial structural similarity to the theoretical optima. This indicates that, even when the optimal policy is highly non-trivial, as in the biased case in figure 2a, the LLM planner exhibits sophisticated strategic behavior that accounts for and capitalizes on social learning.

In the altruistic case, both planners stop investing when public belief is strong and invest heavily when it is uncertain. In the biased case, both planners exhibit the same qualitative pattern: high investment to escape an unfavorable belief, reduced precision near the midpoint, and no investment once a favorable belief has been established. As shown in figure 2b, the magnitude of policy deviation is often modest; for both altruistic and biased planners, deviations are below 10% for the majority of belief states, underscoring broad structural alignment between the emergent LLM strategy and the theoretical characterization.

Nevertheless, several structural differences remain, and these are most naturally interpreted as strategic adaptations to the non-Bayesian behaviors identified in Section 6.1. (1) The LLM planner tends to avoid extreme precisions, such as 0.5 or 1.0, consistent with known central-tendency bias (Rupprecht et al., 2025). (2) The LLM planner’s policy tapers investment more gradually rather than imposing a sharp cutoff. This pattern responds directly to agents’ resistance to cascades (NB3): the planner appears to infer that it is never entirely “safe” to stop investing, because even agents with strong priors can be moved by a countervailing private signal. (3) Similarly, the biased LLM planner continues to invest at very low beliefs, reflecting an apparent recognition that agents may overreact (NB2) to a surprisingly positive signal, thereby making a “last-ditch effort” more viable than in the Bayesian benchmark.

6.3 WELFARE IMPLICATIONS OF ALTRUISM, BIAS, AND ALIGNMENT

We define social welfare $W^\pi(b)$ as the total expected discounted utility of all agents. Because better information never reduces an agent’s expected utility, social welfare is monotone in signal precision (see Appendix C.9), implying that the altruistic planner always increases social welfare relative to the baseline. To quantify these effects, we compare social welfare and planner expenditure across three settings: (1) the *analytical* setting, with an optimal policy and Bayesian agents; (2) the *LLM* setting, with an LLM planner and LLM agents; and (3) the *hybrid* setting, with the optimal policy applied to LLM agents.

The results in figure 2c confirm that planners in all settings can substantially alter social welfare. Moreover, ignoring social learning, as in the myopic cases, substantially worsens planner outcomes. In particular, when misaligned, the biased analytical and LLM planners reduce social welfare by 40 to 50%. They do so by intentionally obscuring information about the true state through policies that decrease signal precision. The magnitude of this welfare loss is especially striking given the stringent constraints imposed on the planners (see Remark 2), and it further substantiates the risks posed by potentially misaligned LLM information mediators.

The results also highlight the cost of agent-model misspecification. In the hybrid setting, the analytically optimal policy, which is designed for Bayes-rational agents, is comparatively “brittle” and performs worse when applied to non-Bayesian agents. The LLM policy, by contrast, closely resembles the analytical policy while being better adapted to non-Bayesian agents with human-like biases.

7 CONCLUSION

We introduced a framework for studying the strategic control of social learning by an algorithmic information mediator and developed a formal model of this increasingly consequential dynamic. The analysis shows that even a tightly constrained planner can substantially affect social welfare, either beneficially or adversely, and that accounting for social learning is essential to exercising such influence effectively. The simulations corroborate these insights, indicating that LLMs are capable of strategic reasoning that resembles the structure of our analytical results while also adapting to non-Bayesian agents.

Taken together, these findings yield several implications for algorithmic information mediation: (1) human-AI and human-algorithm interactions cannot be adequately studied in isolation, because neglecting their interaction with other social processes, including social learning, obscures their positive and negative potential; (2) information mediators, even when exercising comparatively subtle forms of influence, possess substantial capacity to guide or derail social learning and public opinion, which warrants further study of risk-mitigation mechanisms; and (3) modern LLMs display emergent strategic behavior that can exploit both social learning and non-Bayesian cognitive biases. The latter two points underscore the need to better understand how risks associated with algorithmic information mediators, and LLMs in particular, can be mitigated. A central limitation of our study is the paucity of human data. Because the fidelity of LLM-human simulators remains contested (see, e.g., Horton (2023); Gao et al. (2025)), human data would be valuable for validating, and potentially fine-tuning, the LLM simulators.

We close by emphasizing two future directions: (1) generalizing the model and results, and (2) mitigating the negative welfare effects generated by biased planners. Several natural generalizations would relax the assumptions of the present model. Larger state spaces, richer signal structures, more general loss functions, and heterogeneous agents would all bring the framework closer to realistic information environments. The main difficulty in moving beyond binary states and 0-1 losses appears primarily algebraic, and we conjecture that the qualitative structure of the results will persist. For more general signal structures, the central challenge is preserving the convexity result of Theorem 2; identifying the broadest class of signal structures for which such convexity obtains is therefore an especially important problem for future work. As an initial step toward heterogeneity, Appendix D generalizes the altruistic-planner results to agents with opposing preferences.

A second important direction is the design of interventions that prevent welfare losses caused by biased planners. This includes regulatory or mechanism-design approaches that align planner and agent incentives sufficiently to avoid detrimental welfare consequences.

REFERENCES

- Shipra Agrawal, Yiding Feng, and Wei Tang. Dynamic Pricing and Learning with Bayesian Persuasion. *Advances in Neural Information Processing Systems*, 36:59273–59285, December 2023. URL https://proceedings.neurips.cc/paper_files/paper/2023/hash/b9c2e8a0bbcd5fcfaf62856a3a719ada-Abstract-Conference.html.
- Lisa P. Argyle, Christopher A. Bail, Ethan C. Busby, Joshua R. Gubler, Thomas Howe, Christopher Rytting, Taylor Sorensen, and David Wingate. Leveraging AI for democratic discourse: Chat interventions can improve online political conversations at scale. *Proceedings of the National Academy of Sciences*, 120(41):e2311627120, October 2023. ISSN 0027-8424, 1091-6490. doi: 10.1073/pnas.2311627120. URL <https://pnas.org/doi/10.1073/pnas.2311627120>.
- Itai Arieli and Manuel Mueller-Frank. A general analysis of sequential social learning. *Mathematics of Operations Research*, 46:1235–1249, 11 2021.
- Itai Arieli, Ronen Gradwohl, and Rann Smorodinsky. Herd Design. In *Proceedings of the 23rd ACM Conference on Economics and Computation*, pp. 871–872, Boulder CO USA, July 2022. ACM. ISBN 978-1-4503-9150-4. doi: 10.1145/3490486.3538292. URL <https://dl.acm.org/doi/10.1145/3490486.3538292>.
- Cuimin Ba, J. Aislinn Bohren, and Alex Imas. Over- and Underreaction to Information. *SSRN Electronic Journal*, 2022. ISSN 1556-5068. doi: 10.2139/ssrn.4274617. URL <https://www.ssrn.com/abstract=4274617>. Publisher: Elsevier BV.

- Francesco Bacchiocchi, Matteo Bollini, Matteo Castiglioni, Alberto Marchesi, and Nicola Gatti. Online Bayesian Persuasion Without a Clue. *Advances in Neural Information Processing Systems*, 37:76404–76449, December 2024.
- Fengshuo Bai, Mingzhi Wang, Zhaowei Zhang, Boyuan Chen, Yinda Xu, Ying Wen, and Yaodong Yang. Efficient Model-agnostic Alignment via Bayesian Persuasion, May 2024.
- Abhijit V Banerjee. A simple model of herd behavior. *The Quarterly Journal of Economics*, 107(3):797–817, 1992.
- Larry M. Bartels. *Presidential primaries and the dynamics of public choice*. Princeton, N.J. : Princeton University Press, 1988.
- Chris T Bauch and Samit Bhattacharyya. Evolutionary game theory and social learning can determine how vaccine scares unfold. *PLoS computational biology*, 8(4), 2012.
- Dirk Bergemann and Stephen Morris. Information Design: A Unified Perspective. *Journal of Economic Literature*, 57(1):44–95, March 2019. ISSN 0022-0515. doi: 10.1257/jel.20181489. URL <https://www.aeaweb.org/articles?id=10.1257/jel.20181489>.
- Martino Bernasconi, Matteo Castiglioni, Alberto Marchesi, and Mirco Mutti. Persuading Farsighted Receivers in MDPs: The Power of Honesty. *Advances in Neural Information Processing Systems*, 36:14987–15014, December 2023.
- Sujay Bhatt and Vikram Krishnamurthy. Controlled Sequential Information Fusion With Social Sensors. *IEEE Transactions on Automatic Control*, 66(12):5893–5908, December 2021.
- Sushil Bikhchandani, David Hirshleifer, and Ivo Welch. A theory of fads, fashion, custom, and cultural change as informational cascades. *Journal of Political Economy*, 100(5):992–1026, 1992.
- Sushil Bikhchandani, David Hirshleifer, Omer Tamuz, and Ivo Welch. Information Cascades and Social Learning. *Journal of Economic Literature*, 62(3):1040–1093, September 2024. ISSN 0022-0515, 2328-8175. doi: 10.1257/jel.20241472. URL <https://pubs.aeaweb.org/doi/10.1257/jel.20241472>.
- Ilai Bistriz, Nasim Heydaribeni, and Achilleas Anastasopoulos. Informational cascades with nonmyopic agents. *IEEE Transactions on Automatic Control*, 67, 9 2022.
- William J Brady, Killian McLoughlin, Tuan N Doan, and Molly J Crockett. How social learning amplifies moral outrage expression in online social networks. *Science Advances*, 7(33), 2021.
- Carlos Carrasco-Farre. Large Language Models are as persuasive as humans, but how? About the cognitive effort and moral-emotional language of LLM arguments, April 2024. URL <http://arxiv.org/abs/2404.09329>. arXiv:2404.09329 [cs].
- Matteo Castiglioni, Andrea Celli, Alberto Marchesi, and Nicola Gatti. Online Bayesian Persuasion. In *Advances in Neural Information Processing Systems*, volume 33, pp. 16188–16198. Curran Associates, Inc., 2020. URL <https://proceedings.neurips.cc/paper/2020/hash/ba5451d3c91a0f982f103cdbc249bc78-Abstract.html>.
- Matteo Castiglioni, Alberto Marchesi, Andrea Celli, and Nicola Gatti. Multi-Receiver Online Bayesian Persuasion. In *Proceedings of the 38th International Conference on Machine Learning*, pp. 1314–1323. PMLR, July 2021.
- Kenneth Chan, Gary Charness, Chetan Dave, and J. Lucas Reddinger. On Prior Confidence and Belief Updating, May 2025. URL <http://arxiv.org/abs/2412.10662>. arXiv:2412.10662 [econ].
- Zirui Cheng and Jiaxuan You. Towards Strategic Persuasion with Language Models, September 2025.
- Krishna Dasaratha and Kevin He. Learning from viral content. *arXiv preprint arXiv:2210.01267*, 2022.
- Danica Dillion, Niket Tandon, Yuling Gu, and Kurt Gray. Can AI language models replace human participants? *Trends in Cognitive Sciences*, 27(7):597–600, July 2023. ISSN 1364-6613, 1879-307X. doi: 10.1016/j.tics.2023.04.008. URL [https://www.cell.com/trends/cognitive-sciences/abstract/S1364-6613\(23\)00098-0](https://www.cell.com/trends/cognitive-sciences/abstract/S1364-6613(23)00098-0). Publisher: Elsevier.
- Paul Duetting, Safwan Hossain, Tao Lin, Renato Paes Leme, Sai Srivatsa Ravindranath, Haifeng Xu, and Song Zuo. Information Design With Large Language Models, September 2025. URL <http://arxiv.org/abs/2509.25565>. arXiv:2509.25565 [cs].

- Maxim Egorov, Zachary N. Sunberg, Edward Balaban, Tim A. Wheeler, Jayesh K. Gupta, and Mykel J. Kochenderfer. POMDPs.jl: A framework for sequential decision making under uncertainty. *Journal of Machine Learning Research*, 18(26):1–5, 2017.
- Erik Eyster and Matthew Rabin. Naïve herding in rich-information settings. *American Economic Journal: Microeconomics*, 2:221–243, 11 2010.
- Shen Gao, Jiabao Fang, Quan Tu, Zhitao Yao, Zhumin Chen, Pengjie Ren, and Zhaochun Ren. Generative News Recommendation. In *Proceedings of the ACM Web Conference 2024, WWW '24*, pp. 3444–3453, New York, NY, USA, May 2024. Association for Computing Machinery. ISBN 979-8-4007-0171-9. doi: 10.1145/3589334.3645448. URL <https://dl.acm.org/doi/10.1145/3589334.3645448>.
- Yuan Gao, Dokyun Lee, Gordon Burtch, and Sina Fazelpour. Take Caution in Using LLMs as Human Surrogates: Scylla Ex Machina, January 2025. URL <http://arxiv.org/abs/2410.19599>. arXiv:2410.19599 [econ].
- Kenneth T Gillingham and Bryan Bollinger. Social learning and solar photovoltaic adoption. *Management Science*, 67(11):7091–7112, 2021.
- Douglas Guilbeault, Joshua Becker, and Damon Centola. Social learning and partisan bias in the interpretation of climate trends. *Proceedings of the National Academy of Sciences*, 115(39):9714–9719, 2018.
- Keegan Harris, Nicole Immorlica, Brendan Lucier, and Aleksandrs Slivkins. Algorithmic Persuasion Through Simulation, February 2025. URL <http://arxiv.org/abs/2311.18138>. arXiv:2311.18138 [cs].
- Claudia E. Haupt and Mason Marks. AI-Generated Medical Advice—GPT and Beyond. *JAMA*, 329(16): 1349–1350, April 2023. ISSN 0098-7484. doi: 10.1001/jama.2023.5321. URL <https://doi.org/10.1001/jama.2023.5321>.
- K. Hinderer. Lipschitz Continuity of Value Functions in Markovian Decision Processes. *Mathematical Methods of Operations Research*, 62(1):3–22, September 2005.
- John J. Horton. Large Language Models as Simulated Economic Agents: What Can We Learn from Homo Silicus?, April 2023. URL <https://www.nber.org/papers/w31122>.
- Bright Huo, Amy Boyle, Nana Marfo, Wimonchat Tangamornsuksan, Jeremy P. Steen, Tyler McKechnie, Yung Lee, Julio Mayol, Stavros A. Antoniou, Arun James Thirunavukarasu, Stephanie Sanger, Karim Ramji, and Gordon Guyatt. Large Language Models for Chatbot Health Advice Studies: A Systematic Review. *JAMA Network Open*, 8(2):e2457879, February 2025. ISSN 2574-3805. doi: 10.1001/jamanetworkopen.2024.57879. URL <https://doi.org/10.1001/jamanetworkopen.2024.57879>.
- Cameron R. Jones and Benjamin K. Bergen. Lies, Damned Lies, and Distributional Language Statistics: Persuasion and Deception with Large Language Models, December 2024. URL <http://arxiv.org/abs/2412.17128>. arXiv:2412.17128 [cs].
- Lodewijk Kallenberg. Markov decision processes. *Lecture Notes. University of Leiden*, 428, 2011.
- Emir Kamenica and Matthew Gentzkow. Bayesian persuasion. *American Economic Review*, 101(6):2590–2615, 2011.
- Vikram Krishnamurthy. Quickest Detection POMDPs With Social Learning: Interaction of Local and Global Decision Makers. *IEEE Transactions on Information Theory*, 58(8):5563–5587, August 2012.
- LG AI Research. Exaone deep: Reasoning enhanced language models. *arXiv preprint arXiv:2503.12524*, 2025.
- Minqian Liu, Zhiyang Xu, Xinyi Zhang, Heajun An, Sarvech Qadir, Qi Zhang, Pamela J. Wisniewski, Jin-Hee Cho, Sang Won Lee, Ruoxi Jia, and Lifu Huang. LLM Can be a Dangerous Persuader: Empirical Study of Persuasion Safety in Large Language Models, April 2025. URL <http://arxiv.org/abs/2504.10430>. arXiv:2504.10430 [cs].
- Nunzio Lorè and Babak Heydari. Strategic behavior of large language models and the role of game structure versus contextual framing. *Scientific Reports*, 14(1), August 2024. ISSN 2045-2322. doi: 10.1038/s41598-024-69032-z. URL <https://www.nature.com/articles/s41598-024-69032-z>. Publisher: Springer Science and Business Media LLC.
- Yaw Nyarko. On the convexity of the value function in Bayesian optimal control problems. *Economic Theory*, 4(2):303–309, March 1994.

- Kenneth Payne and Baptiste Alloui-Cros. Strategic Intelligence in Large Language Models: Evidence from evolutionary Game Theory, July 2025. URL <http://arxiv.org/abs/2507.02618>. arXiv:2507.02618 [cs].
- Yujin Potter, Shiyang Lai, Junsol Kim, James Evans, and Dawn Song. Hidden Persuaders: LLMs’ Political Leaning and Their Influence on Voters, November 2024.
- Martin L. Puterman. Chapter 8 markov decision processes. In *Stochastic Models*, volume 2 of *Handbooks in Operations Research and Management Science*. Elsevier, 1990.
- Maya Raifer, Guy Rotman, Reut Apel, Moshe Tennenholtz, and Roi Reichart. Designing an Automatic Agent for Repeated Language-based Persuasion Games. *Transactions of the Association for Computational Linguistics*, 10:307–324, March 2022.
- Neel Rao, Markus M Mobius, and Tanya Rosenblat. Social networks and vaccination decisions. *FRB of Boston*, 2007.
- Alexander Rogiers, Sander Noels, Maarten Buyl, and Tijl De Bie. Persuasion with Large Language Models: a Survey, November 2024. URL <http://arxiv.org/abs/2411.06837>. arXiv:2411.06837 [cs].
- Jens Rupperecht, Georg Ahnert, and Markus Strohmaier. Prompt Perturbations Reveal Human-Like Biases in LLM Survey Responses, July 2025. URL <http://arxiv.org/abs/2507.07188>. arXiv:2507.07188 [cs].
- Dr.-Ing. Martin Schiele, Yannick Gittmann, Stefan Ilchmann, Ante Gojsalić, Dominik Jurinčić, and Phylis Klempt. Voting Advice Applications: Implementation of RAG-supported LLMs, July 2024. URL <https://www.techrxiv.org/users/802685/articles/1187651-voting-advice-applications-implementation-of-rag-supported-llms?commit=c540a54749906f8ac761093f5dff0f514291d4b6>.
- James E. Smith and Kevin F. McCardle. Structural Properties of Stochastic Dynamic Programs. *Operations Research*, 50(5):796–809, October 2002.
- Lones Smith and Peter Sørensen. Pathological outcomes of observational learning. *Econometrica*, 68:371–398, 2000.
- Lones Smith, Peter Norman Sørensen, and Jianrong Tian. Informational Herding, Optimal Experimentation, and Contrarianism. *The Review of Economic Studies*, 88(5):2527–2554, September 2021. ISSN 0034-6527, 1467-937X. doi: 10.1093/restud/rdab001. URL <https://academic.oup.com/restud/article/88/5/2527/6149316>.
- Shih-Tang Su, Vijay G. Subramanian, and Grant Schoenebeck. Bayesian Persuasion in Sequential Trials. In Michal Feldman, Hu Fu, and Inbal Talgam-Cohen (eds.), *Web and Internet Economics*, pp. 22–40, Cham, 2022. Springer International Publishing.
- Jeremy G Weber. Social learning and technology adoption: The case of coffee pruning in peru. *Agricultural Economics*, 43:73–84, 2012.
- Xupeng Wei and Achilleas Anastasopoulos. Social learning with a self-interested coordinator. In *2022 IEEE 61st Conference on Decision and Control (CDC)*, pp. 3137–3144. IEEE, 2022.
- Marcus Williams, Micah Carroll, Adhyayan Narang, Constantin Weissner, Brendan Murphy, and Anca Dragan. On Targeted Manipulation and Deception when Optimizing LLMs for User Feedback, February 2025. URL <http://arxiv.org/abs/2411.02306>. arXiv:2411.02306 [cs].
- Jibang Wu, Zixuan Zhang, Zhe Feng, Zhaoran Wang, Zhuoran Yang, Michael I. Jordan, and Haifeng Xu. Sequential Information Design: Markov Persuasion Process and Its Efficient Reinforcement Learning, February 2022.
- Shuo Wu, Pawan Poojary, and Randall Berry. Observational Learning with a Budget, April 2025. URL <http://arxiv.org/abs/2504.19396>. arXiv:2504.19396 [cs].
- Tianyi Zhang, Faisal Ladhak, Esin Durmus, Percy Liang, Kathleen McKeown, and Tatsunori B. Hashimoto. Benchmarking Large Language Models for News Summarization. *Transactions of the Association for Computational Linguistics*, 12:39–57, January 2024a. ISSN 2307-387X. doi: 10.1162/tacl_a_00632. URL https://doi.org/10.1162/tacl_a_00632.
- Yadong Zhang, Shaoguang Mao, Tao Ge, Xun Wang, Adrian de Wynter, Yan Xia, Wenshan Wu, Ting Song, Man Lan, and Furu Wei. LLM as a Mastermind: A Survey of Strategic Reasoning with Large Language Models, April 2024b. URL <http://arxiv.org/abs/2404.01230>. arXiv:2404.01230 [cs].

A APPLICATIONS ELABORATED

Consider a sequence of individuals, or agents in our terminology, deciding whether to patronize a business or service provider, such as a restaurant, contractor, or realtor, while a recommendation system, the planner, supplies information about that business. The binary true state of the world is whether the business is good, and this state is initially unknown. Both the recommendation system and the agents seek consistency with the true state: agents want to patronize the business if and only if it is good, and the recommendation system wants to induce precisely that behavior. In this sense, the recommendation system is an *altruistic planner*.

The recommendation system may show an agent a highly targeted ad—*precise* in our terminology—that presents the business’s strengths and weaknesses in contexts intelligible to her background and characteristics, or it may instead show a generic or confusing ad that provides little decision-relevant information. A precise signal is more informative, comprehensible, and relatable for the agent, and is therefore more likely to lead her to the correct conclusion, namely patronizing the business if and only if it is good. Social welfare rises as more agents make correct decisions. Nevertheless, altering ad precision is costly because effective tailoring requires information about the agent’s specific background; hence the recommendation system must choose precisions to maximize social welfare net of tailoring costs.

We next provide a real-world example of a *biased planner*. Consider voters, again modeled as agents, who are motivated to support the candidate most likely to win in their community. The desire to “back a winner” has been shown to affect electoral outcomes; for instance, it helps explain why US states with early primary elections can exert outsized influence on election results (see Bartels (1988)). The planner is a particular candidate’s campaign, which seeks to motivate agents to support that candidate. The binary true state indicates whether the candidate is winning. Thus, if an agent knew the true state, she would support the candidate if and only if he were winning.

Each agent understands her community well enough that, if she accurately understood the candidate’s stances and policies, she would know whether the candidate is winning, i.e., the true state of the world. Accurate understanding, however, depends on presenting the information in a form she can process effectively: some agents respond best to audiovisual material, others to long-form articles supported by facts, figures, and citations, others to brief focused content, and still others to communication in their native language.

The campaign sends digital advertisements, interpreted as private signals, to agents at varying levels of precision. Precision represents the degree to which the content is tailored to the agent’s communicative preferences. A precise signal accurately conveys the true state by helping the agent understand the candidate’s policies, record, and character, thereby enabling her to infer correctly whether the candidate will win. Such precision does not necessarily benefit the planner: if the candidate is in fact losing in the agent’s community, a highly precise signal would induce opposition. An imprecise signal, by contrast, may be more likely to induce support because it obfuscates the true state, perhaps through vagueness or confusion, thereby increasing the probability that the agent believes the candidate is winning. The biased planner may therefore have an incentive to decrease signal precision when it believes its candidate is losing. Tailoring any ad requires research into how the agent best processes content, and this imposes costs; even making a signal highly imprecise can require personalization, for example knowing that an agent who prefers brief focused messages can be confused by verbose long-form material. The planner consequently selects precisions to maximize the expected number of supporters net of precision-generation costs.

B SOCIAL LEARNING DYNAMICS

We first summarize the sequence of events at step i :

1. The planner determines q_i based on $\mathcal{H}_i$ and, q_i is observed by all agents.
2. Agent i receives private signal s_i such that $\mathbb{P}\{s_i = \omega\} = q_i$.
3. Agent i takes action a_i based on the observable history and their private signal with the aim of maximizing their own utility.
4. All other agents (and the planner) observe a_i and the public belief at the end of step i , b_{i+1} is updated accordingly.

Because step (1) makes q_i a function of $\mathcal{H}_i$, conditioning a random variable on both q_i and $\mathcal{H}_i$ renders the conditioning on q_i redundant. We use this observation throughout.

B.1 PRIVATE BELIEF UPDATE

We now derive agent i ’s action rule. Let $\alpha_i := \frac{b_i}{1-b_i}$ denote the public likelihood ratio implied by observed actions. Agent i combines the current public belief with her private signal s_i to form the private belief $\tilde{b}_i =$

$\mathbb{P}\{\omega = G|\mathcal{H}_i, q_i, s_i\}$, and we write the corresponding private likelihood ratio as $\tilde{\alpha}_i := \frac{\tilde{b}_i}{1-\tilde{b}_i}$. The computation of $\tilde{\alpha}_i$ and $\tilde{b}_i$ proceeds as follows.

1. If private signal is $s_i = G$, then

$$\begin{aligned}\tilde{\alpha}_i &= \frac{\tilde{b}_i}{1-\tilde{b}_i} = \frac{\mathbb{P}\{\omega = G|\mathcal{H}_i, q_i, s_i = G\}}{\mathbb{P}\{\omega = B|\mathcal{H}_i, q_i, s_i = G\}} \\ &= \frac{\mathbb{P}\{s_i = G|\omega = G, \mathcal{H}_i, q_i\} \mathbb{P}\{\omega = G|\mathcal{H}_i, q_i\}}{\mathbb{P}\{s_i = G|\omega = B, \mathcal{H}_i, q_i\} \mathbb{P}\{\omega = B|\mathcal{H}_i, q_i\}} \\ &= \frac{q_i}{1-q_i} \frac{b_i}{1-b_i} \tag{8}\end{aligned}$$

$$= \frac{q_i}{1-q_i} \alpha_i \tag{9}$$

In obtaining Equation (8), we use the conditional-independence assumption: given ω , agents' signals are independent of one another and of the history, so $\mathbb{P}\{s_i = G|\omega = G, \mathcal{H}_i\} = \mathbb{P}\{s_i = G|\omega = G\} = q_i$. The same argument applies under conditioning on $\omega = B$, and we also use that q_i is a function of $\mathcal{H}_i$.

2. If the private signal is $s_i = B$, the same logic yields

$$\tilde{\alpha}_i = \frac{1-q_i}{q_i} \alpha_i \tag{10}$$

Let $y(b, q)$ and $z(b, q)$ denote the probabilities of observing signals G and B , respectively, given public belief b and signal precision q conditional on the history. Conditioning on ω and applying the Law of Total Probability gives:

$$\begin{aligned}y(b_i, q_i) &= \mathbb{P}\{s_i = G|\mathcal{H}_i\} \\ &= \sum_{k \in \{G, B\}} \mathbb{P}\{s_i = G|\mathcal{H}_i, \omega = k\} \mathbb{P}\{\omega = k|\mathcal{H}_i\} \\ &= q_i b_i + (1-q_i)(1-b_i) = 1 + 2q_i b_i - q_i - b_i \tag{11}\end{aligned}$$

Similarly,

$$z(b_i, q_i) = q_i + b_i - 2q_i b_i \tag{12}$$

Combining Equation (9), Equation (10), Equation (11), and Equation (12) yields:

$$\tilde{b}_i = \mathbb{P}(\omega = G|\mathcal{H}_i, q_i, s_i) = f(b_i, q_i, s_i) = \begin{cases} \frac{q_i}{y(b_i, q_i)} b_i & s_i = G \\ \frac{1-q_i}{z(b_i, q_i)} b_i & s_i = B \end{cases} \tag{13}$$

If $s_i = G$, then $y(b_i, q_i) = \mathbb{P}(s_i = G|\mathcal{H}_i)$ cannot equal 0; similarly, if $s_i = B$, then $z(b_i, q_i)$ cannot equal 0. The expression above is therefore well-defined.

B.2 AGENT ACTIONS AND PUBLIC BELIEF

Agent i 's action a_i is G whenever $\tilde{b}_i$ exceeds 0.5 (equivalently, $\tilde{\alpha}_i > 1$), is B whenever $\tilde{b}_i < 0.5$ (equivalently, $\tilde{\alpha}_i < 1$), and is s_i under the specified tie-breaking rule. This action, together with agent i 's precision q_i , is observed by all peers and thereby determines the subsequent public-belief update.

When $b_i < 1 - q_i$, we have $\alpha_i < \frac{1-q_i}{q_i}$. Hence, if $s_i = G$, then $\tilde{\alpha}_i < 1$ by Equation (9); if $s_i = B$, then $\tilde{\alpha}_i < \frac{(1-q_i)^2}{q_i^2} \leq 1$ (because $q_i \geq 0.5$) by Equation (10). Thus, irrespective of the private-signal realization s_i , the agent chooses $a_i = B$ whenever $b_i < 1 - q_i$. A symmetric argument shows that $a_i = G$ whenever $b_i > q_i$. Intuitively, these are precisely the regions in which the public belief is sufficiently strong to dominate the private signal's precision.

Now consider the intermediate region $1 - q_i \leq b_i \leq q_i$. If $s_i = G$, then Equation (8) and $b_i \geq 1 - q_i$ imply $\tilde{\alpha}_i \geq 1$; if $s_i = B$, then Equation (10) and $b_i \leq q_i$ imply $\tilde{\alpha}_i \leq 1$. Together with the tie-breaking rule, these inequalities yield $a_i = s_i$, and hence Equation (2), reproduced below.

$$a_i = \begin{cases} s_i & 1 - q_i \leq b_i \leq q_i \\ G & q_i < b_i \\ B & q_i < 1 - b_i \end{cases}$$

Consequently, agent i 's action is informative only in the first case of Equation (2), where it permits future agents to infer i 's private signal perfectly. In that case, the public belief updates exactly as the private belief in Equation (13). In the latter two cases of Equation (2), the action contains no incremental information about the true state, so the public belief remains unchanged. This gives Equation (3), reproduced below:

$$b_{i+1} = f(b_i, q_i) = \begin{cases} \tilde{b}_i & 1 - q_i \leq b_i \leq q_i \\ b_i & \text{o.w.} \end{cases} \quad (14)$$

B.3 MARKOVIANITY AND SUFFICIENCY OF PUBLIC BELIEF

We now show that $\mathbb{P}\{a_i = a | \mathcal{H}_i, q_i\}$ depends on the history only through b_i and q_i , for each $a \in \{G, B\}$. It follows that

$$\mathbb{P}(a_i = a | \mathcal{H}_i, q_i) = \mathbb{P}(a_i = a | b_i, q_i) \quad (15)$$

In the second and third cases of Equation (2), the claim is immediate because the action is pinned down by b_i and q_i alone. It remains to consider the first case.

$$\begin{aligned} \mathbb{P}(a_i = G | \mathcal{H}_i, q_i) &= \sum_{\omega \in \{G, B\}} \mathbb{P}(\omega = \omega | \mathcal{H}_i, q_i) \mathbb{P}(a_i = G | \mathcal{H}_i, q_i, \omega = \omega) \\ &= b_i \mathbb{P}(a_i = G | \mathcal{H}_i, q_i, \omega = G) \\ &\quad + (1 - b_i) \mathbb{P}(a_i = G | \mathcal{H}_i, q_i, \omega = B) \\ &= b_i \mathbb{P}(s_i = G | b_i, q_i, \omega = G) \\ &\quad + (1 - b_i) \mathbb{P}(s_i = G | b_i, q_i, \omega = B) \\ &= b_i q_i + (1 - b_i)(1 - q_i) \end{aligned} \quad (16)$$

The first equality applies the law of total probability. The second uses the fact that q_i is a function of $\mathcal{H}_i$, allowing substitution of b_i for $\mathbb{P}(\omega = w | \mathcal{H}_i, q_i)$. The third uses $a_i = s_i$ in the region $1 - q_i \leq b_i \leq q_i$, together with conditional independence of s_i from the history given ω . The result follows.

This establishes the following theorem:

Theorem 6. *Markovianity of the Public Belief Process*

$$b_{i+1} = \mathbb{P}\{\omega = G | b_i, q_i, a_i\} \quad (17)$$

Equivalently, b_i contains all information in $\mathcal{H}_i$ that is relevant for updating b_{i+1} .

Proof. Note that, from step (1) in Appendix B, q_i is a function of $\mathcal{H}_i$. It suffices to show

$$b_{i+1} = \frac{\mathbb{P}\{a_i \cap \omega = G | b_i, q_i\}}{\mathbb{P}\{a_i | b_i, q_i\}}$$

We will prove the claim via induction on i .

For our base case, consider $i = 1$ so that $\mathcal{H}_2 = \{b_1, (q_1, a_1)\}$ and $b_i = b_1$. An application of Bayes' Theorem provides the following:

$$\begin{aligned} b_2 &= \frac{\mathbb{P}\{\omega = G | b_1, q_1\} \mathbb{P}\{a_1 | \omega = G, b_1, q_1\}}{\mathbb{P}\{a_1 | b_1, q_1\}} = \frac{b_1 \mathbb{P}\{a_1 | \omega = G, b_1, q_1\}}{\mathbb{P}\{a_1 | b_1, q_1\}} \\ &= \frac{b_1 \mathbb{P}\{a_1 \cap \omega = G | b_1, q_1\}}{\mathbb{P}\{a_1 | b_1, q_1\} \mathbb{P}\{\omega = G | b_1, q_1\}} \\ &= \frac{\mathbb{P}\{a_1 \cap \omega = G | b_1, q_1\}}{\mathbb{P}\{a_1 | b_1, q_1\}} \end{aligned}$$

where the final step follows as the unconditioned belief of ω is equal to the prior belief b_1 .

Now, assume the claim holds for all $i \leq n$.

$$\begin{aligned} b_{n+1} &= \mathbb{P}\{\omega = G | \mathcal{H}_n, a_n, q_n\} = \frac{\mathbb{P}\{\omega = G | \mathcal{H}_n, q_n\} \mathbb{P}\{a_n | \omega = G, \mathcal{H}_n, q_n\}}{\mathbb{P}\{a_n | \mathcal{H}_n, q_n\}} \\ &= \frac{b_n \mathbb{P}\{a_n | \omega = G, \mathcal{H}_n, q_n\}}{\mathbb{P}\{a_n | \mathcal{H}_n, q_n\}} \\ &= \frac{b_n \mathbb{P}\{a_n \cap \omega = G | \mathcal{H}_n, q_n\}}{\mathbb{P}\{a_n | b_n, q_n\} \mathbb{P}\{\omega = G | \mathcal{H}_n, q_n\}} \\ &= \frac{\mathbb{P}\{a_n \cap \omega = G | b_n, q_n\}}{\mathbb{P}\{a_n | b_n, q_n\}} \end{aligned}$$

where the final two steps follow from Equation (15) and the definition of b_n .

This completes the proof. $\square$

Note that we have shown that the public belief is Markovian adapting the arguments in Banerjee (1992) and Bikhchandani et al. (1992). The modifications were required because of the differences in selections of the agents' precisions caused by the introduction of the central planner.

C PROOFS

C.1 DERIVATION OF EQUATION (4)

Lemma 7.

$$\mathbb{P}(a_i \neq \omega | b_i, q_i) = \min(b_i, 1 - b_i, 1 - q_i)$$

Proof. We prove this for each case of a_i delineated in Equation (2). In the second case, we can reason as follows:

$$\mathbb{P}(a_i \neq \omega | b_i, q_i) = \mathbb{P}(\omega \neq G | b_i, q_i) = 1 - b_i$$

Similarly, in the third case of Equation (2), we obtain:

$$\mathbb{P}(a_i \neq \omega | b_i, q_i) = \mathbb{P}(\omega \neq B | b_i, q_i) = b_i$$

For the case when $1 - q_i \leq b_i \leq q_i$ and $a_i = s_i$ from Equation (2), we can reason as follows:

$$\begin{aligned} \mathbb{P}(a_i \neq \omega | b_i, q_i) &= \sum_{a \in \{G, B\}} \mathbb{P}(a_i = a | b_i, q_i) \mathbb{P}(\omega \neq a | b_i, q_i, a_i = a) \\ &= \mathbb{P}(s_i = G | b_i, q_i) \mathbb{P}(\omega \neq G | b_i, q_i, s_i = G) \\ &\quad + \mathbb{P}(s_i = B | b_i, q_i) \mathbb{P}(\omega \neq B | b_i, q_i, s_i = B) \end{aligned} \tag{Eqn 2} \tag{18}$$

$$\begin{aligned} &= y(b_i, q_i) \mathbb{P}(\omega = B | b_i, q_i, s_i = G) \\ &\quad + z(b_i, q_i) \mathbb{P}(\omega = G | b_i, q_i, s_i = B) \end{aligned} \tag{Eqns 11, 12, Thm 6} \tag{19}$$

$$= \begin{cases} y(0, q_i) & b_i = 0 \\ y(b_i, q_i)(1 - \frac{b_i q_i}{y(b_i, q_i)}) + z(b_i, q_i) \frac{b_i(1 - q_i)}{z(b_i, q_i)} & b_i \neq 0 \end{cases} \tag{Eqn 13} \tag{20}$$

$$= \begin{cases} 1 - q_i & b_i = 0 \\ y(b_i, q_i) - b_i q_i + b_i(1 - q_i) & b_i \neq 0 \end{cases}$$

$$= \begin{cases} 1 - q_i & b_i = 0 \\ y(b_i, q_i) - b_i q_i + b_i - b_i q_i & b_i \neq 0 \end{cases}$$

$$= \begin{cases} 1 - q_i & b_i = 0 \\ 1 - q_i & b_i \neq 0 \end{cases}$$

Combining the three cases yields the claim. $\square$

C.2 PROOF OF THEOREM 1: MYOPIC ALTRUISTIC POLICY

Applying Equation (4) to $r_A(b, q)$ in Equation (5) allows us to reason as follows

$$\frac{\partial r_A(b, q)}{\partial q} = \begin{cases} -\dot{\beta}(q) & q < \max(b, 1 - b) \\ -\dot{\beta}(q) + C & q \geq \max(b, 1 - b) \end{cases} \tag{21}$$

As stated in Section 3.2, we can restrict the altruistic planner to precision $q \in [p, 1]$ because social welfare is increasing in precision (see Appendix C.9). Accordingly, $\beta(p) = 0$.

Because $\beta(\cdot)$ is increasing, the first case of Equation (21) leads to optimal precision p if $p < \max(b, 1 - b)$. That is, if the signal is not precise enough to influence the agent's action, then the planner will not expend resources for it. In the latter case, because $\beta(\cdot)$ is increasing and concave, $\dot{\beta}(\cdot)$ is positive and decreasing. Thus, $r_A(b, q)$, if the optimal lies in this range it must be attained at $q = 1$.

We now identify the regime in which the latter case is optimal.

$$\begin{aligned} r_A(b, p) &< r_A(b, 1) \\ -C \min(b, 1-b, 1-p) &< -\beta(1) \\ \min(b, 1-b, 1-p) &> \frac{\beta(1)}{C} \end{aligned}$$

If $1-p \leq \frac{\beta(1)}{C}$, the above inequality is never satisfied, and it is never optimal to select precision 1. Otherwise, the inequality above yields $b \in \left[\frac{\beta(1)}{C}, 1 - \frac{\beta(1)}{C}\right]$. This yields Theorem 1. ■

C.3 PROOF OF THEOREM 2: ALTRUISTIC VALUE FUNCTION CONVEXITY

Proof Sketch We inductively prove that the expected k -th stage reward, i.e., the expected utility of the planner from the control and action of the k -th agent, is convex. The instantaneous reward (Equation (5)), which is convex with respect to public belief, provides our base case.

The first challenge encountered is the unusual nature of the public belief update. Although the state space is uncountably infinite, the belief update only takes support on a maximum of 2 values. To manage this, we define a decision tree, i.e., the complete binary tree of all possible trajectories once an initial belief and policy are fixed. Each node has two children corresponding to each possible signal realization the next agent might receive. The root is the expected instantaneous reward at time 1, i.e., from the first agent's action. The induction moves down the levels of this tree with the k -th level containing 2^{k-1} nodes, each associated with a sequence of realizations of $k-1$ signals.

We then show that for a node in the $(k-1)$ -th level that has convex expected reward, its two children in the k -th level satisfy the same property. This is where we must deal with the dependence of agents' actions on public belief. Note that even when applying the same precision and receiving the same signal realization, two agents beginning at different public beliefs may take opposing actions (see Equation (2)). Thus, standard results that provide easy ways of bounding the future terms of the Markov process do not apply. Here, our specific belief update is actually helpful. We can leverage the fact that Bayesian updates are martingales (i.e., $\mathbb{E}[b_{i+1}] = b_i$). Along with the convexity of instantaneous rewards, this allows us to complete the inductive step and, subsequently, the proof.

Preliminaries We first prove the following lemma containing a few useful properties of the optimal value function $V_A^*(\cdot)$.

Lemma 8.

$V_A^*(\cdot)$ is non-positive, $V_A^*(b) = V_A^*(1-b) \forall b \in [0, 1]$, and $V_A^*(0) = V_A^*(1) = 0$ with $\pi_A^*(0) = \pi_A^*(1) = p$.

Proof.

Non-positivity

$$r(b_i, q_i) \leq 0 \quad \forall b_i, q_i \rightarrow V_A^* \leq 0$$

Symmetry

Let $n_i = 1 - b_i$ and let Λ_n and Λ_b be the decision trees rooted at states n_0 and b_0 , respectively. Showing that Λ_n and Λ_b are symmetric suffices to show the symmetry of $V^*(\cdot)$. This will be accomplished by showing that the expected rewards, belief state values, and branching probabilities of the trees are reflections of one another.

First consider the instantaneous reward function r_A . Note that the decision function of the agents when choosing their actions is symmetric from Equation (2), thus, $\mathbb{P}\{a_i = G|b_i\} = \mathbb{P}\{a_i = B|n_i\}$. Applying this, we obtain

$$\begin{aligned} r(n_i, q_i) &= -\beta(q_i) - C n_i \mathbb{P}\{a_i = B|\omega = G, n_i\} - C(1 - n_i) \mathbb{P}\{a_i = G|\omega = B, n_i\} \\ &= -\beta(q_i) - C(1 - b_i) \mathbb{P}\{a_i = B|\omega = G, n_i\} - C b_i \mathbb{P}\{a_i = G|\omega = B, n_i\} \\ &= -\beta(q_i) - C(1 - b_i) \mathbb{P}\{a_i = G|\omega = B, b_i\} - C b_i \mathbb{P}\{a_i = B|\omega = G, b_i\} \\ &= r(b_i, q_i) \end{aligned}$$

Thus, the instantaneous reward function is symmetric about 0.5.

Now consider the belief state update when beginning with belief n_i assuming signal precision q_i . we will use $\bar{x}$ and $\underline{x}$ to represent updated beliefs from x after observing action G or B , respectively. From Equation (13):

$$\begin{aligned} \bar{n}_i &= \frac{q_i}{y(n_i, q_i)} n_i = \frac{q_i}{z(b_i, q_i)} (1 - b_i) = 1 - \frac{1 - q_i}{z(b_i, q_i)} b_i = 1 - \underline{b}_i \\ \underline{n}_i &= \frac{1 - q_i}{z(n_i, q_i)} n_i = \frac{q_i}{y(b_i, q_i)} (1 - b_i) = 1 - \frac{q_i}{y(b_i, q_i)} b_i = 1 - \bar{b}_i \end{aligned}$$

Thus, the belief updates are also symmetric. Combining this with the symmetry of the reward we obtain:

$$\begin{aligned} r(\bar{n}_i) &= r(1 - \underline{b}_i) = r(\underline{b}_i) \\ r(\underline{n}_i) &= r(1 - \bar{b}_i) = r(\bar{b}_i) \end{aligned}$$

Thus, the expected rewards and belief state values of Λ_n and Λ_b are mirror images of one another.

Finally, consider the branching probabilities which we will denote as $y(b, q) = 1 + 2bq - b - q$ for observing action G and $z(b, q) = 1 - y(b, q)$ for observing action B .

$$\begin{aligned} y(n, q) &= 1 + 2nq - n - q = 1 + 2(1 - b)q - (1 - b) - q = b + q - 2bq = z(b, q) \\ z(n, q) &= 1 - y(n, q) = 1 - z(b, q) = y(b, q) \end{aligned}$$

Thus, the branching probabilities of Λ_n and Λ_b are also symmetric, completing the proof.

Extremal values

We will show that $V_A^*(0) = 0$ and rely on the symmetry about 0.5 proven above to show that $V_0^*(1) = 0$.

From Equation (3), when $b_0 = 0$, $b_0^+ = b_0^- = 0$ for any value of precision. Thus, regardless of signal realizations or control actions, the agents' actions and, subsequently, the public belief, will remain unchanged. That is, $b_i = 0 \forall i$ and, from Equation (2), $a_i = B \forall i$.

From Equation (5), we can then write:

$$\begin{aligned} V_A^*(0) &= \max_{\pi \in \Pi} \mathbb{E} \left[\sum_{i=0}^{\infty} \delta^i r(b_i, \pi(b_i)) \right] \\ &= \max_{\pi \in \Pi} \mathbb{E} \left[\sum_{i=0}^{\infty} \delta^i r(0, \pi(0)) \right] \\ &= \max_{\pi \in \Pi} \mathbb{E} \left[\sum_{i=0}^{\infty} \delta^i [-\beta(\pi(0)) - C b_i \mathbb{P}\{a_i = B | \omega = G, b_i = 0\} - C(1 - b_i) \mathbb{P}\{a_i = G | \omega = B, b_i = 0\}] \right] \\ &= \max_{\pi \in \Pi} \mathbb{E} \left[\sum_{i=0}^{\infty} \delta^i [-\beta(q_i)] \right] \end{aligned} \tag{22}$$

Thus, to attain $V_A^*(0)$, it suffices to choose $\pi(0)$ such that $\beta(\pi(0))$ is minimized for each i . Recall that $\beta(\cdot)$ is increasing on $[0.5, 1]$ and takes its minimum at $\beta(p) = 0$. Thus, by choosing $q_i = p \forall i$, Equation 22 yields $V_A^*(0) = 0$ with corresponding optimal policy $\pi_A^*(0) = p$.

The aforementioned symmetry about 0.5, shows that $V_A^*(1) = 0$, completing the proof. $\square$

We next introduce *decision trees* for both Markovian and more general policies. A policy can be represented by a complete binary tree, as in figure 3, whose nodes encode system states, namely the belief values at the moments when the controller sends private signals to agents at the corresponding levels. Branches encode the actions taken by those agents after receiving their private signals. The controller's policy, through its precision choices, determines both the action probabilities emanating from each node and the belief values induced by those actions; these action probabilities can be viewed as branch weights. A decision tree is therefore uniquely specified by node-level belief values and branch weights. For a deterministic stationary Markovian policy, the controller's precision choice, and hence each action probability, is a deterministic function only of the node's belief value, independent of both the node's depth and the path from the root. Accordingly, the decision tree associated with any such policy is fully determined by the initial state. Under an arbitrary controller policy, by contrast, the decision at a node may depend on the entire path leading to that node.

Proof of Theorem 2 We now prove the convexity of $V_A^*(\cdot)$ with respect to public belief.

Proof.

We will first show that, for fixed precision, the instantaneous reward r_A is convex in public belief. We can rewrite r_A as follows using Lemma 7:

$$r_A(x, q) = -\beta(q) - C \min(x, 1 - x, 1 - q) \tag{23}$$

The pointwise minimum of linear functions is concave. As we are subtracting the minimum, the overall function is convex with respect to x .

We consider arbitrary x_0, m_0, n_0 such that $tm_0 + (1-t)n_0 = x_0$ for some $t \in [0, 1]$. To show the convexity of $V_A^*(\cdot)$ we must show that $V_A^*(x_0) \leq tV_A^*(m_0) + (1-t)V_A^*(n_0)$. By definition of the optimal value function, $V_A^*(x) \geq V_A^\pi(x)$ for any policy π . Thus, it will suffice to show that there exist $\tilde{\pi}$ such that

$$V_A^*(x_0) \leq tV_A^{\tilde{\pi}}(m_0) + (1-t)V_A^{\tilde{\pi}}(n_0) \quad (24)$$

We refer to the decision tree of π^* when the initial state is x_0 as Λ .

We consider $\tilde{\pi}$ to be a policy of the controller that chooses the precision at each node to be the same as what Λ does at the counterpart node. The nodes of decision trees are considered to be counterparts if they can be reached via the same sequence of actions from the root (see figure 3 for illustration). Thus, the precision chosen by this policy is the same as what Λ does after the same set of actions of the agents. We refer to the decision trees for $\tilde{\pi}$ with initial state m_0 (respectively, n_0) as Λ_m (Λ_n).

We now illustrate the policy $\tilde{\pi}$ for initial values m_0 and n_0 considering the decision trees Λ_m and Λ_n . While doing so, we illustrate counterpart nodes and the weights of the branches. We will select one path of actions λ out of these decision trees by specifying set $\mathcal{G}$ as the epochs where action G is taken and $\mathcal{B}$ as its complement. Refer to figure 3 for illustration. Considering the decision tree Λ , we will refer to x_i as the state at epoch i when beginning with belief x_0 and following π^* along this path of actions. That is, x_i assumes one of 2^i possible belief values at epoch i starting from x_0 . In decision tree Λ_m (respectively, Λ_n), we use m_i (n_i) to denote the public belief after beginning with m_0 (n_0), following the path λ . Node m_i (respectively n_i) in decision tree Λ_m (Λ_n) is the counterpart of node x_i from decision tree Λ . Note that, $\tilde{\pi}(m_i)$ can now be illustrated notationally. Specifically, $\tilde{\pi}(m_i) = \tilde{\pi}(n_i) = \pi_A^*(x_i)$. Similarly, $\tilde{\pi}$ can be defined at each node of Λ_m , considering the action taken by π^* at each counterpart node in Λ . Note that $y(x_i, \pi_A^*(x_i))$ (respectively, $z(x_i, \pi_A^*(x_i))$) is the weight of the branch corresponding to G (B) emanating from x_i in Λ . Because $\tilde{\pi}(m_i) = \tilde{\pi}(n_i) = \pi_A^*(x_i)$, $y(m_i, \tilde{\pi}(m_i)) = y(m_i, \pi_A^*(x_i))$ is the weight of the branch emanating from m_i for action G in Λ_m . This extends similarly to branches associated with action B and Λ_n . Thus, the three decision trees differ in both the values of the nodes and the weights of the branches. Since π^* is deterministic, so is $\tilde{\pi}$. Therefore, from Equation (13) and Equation (3), x_i, m_i , and n_i are deterministic quantities once the specific path of actions is specified by $\mathcal{G}$.

Now let $\phi_k^\pi(b_0) = \mathbb{E}[r_A(b_k, \pi(b_k))]$ be the expected instantaneous reward at epoch k when beginning with state b_0 and applying policy π and $\phi^*(x) = \phi^{\pi^*}(x)$. Applying Fubini's Theorem, we can rewrite our value function as

$$V_A^\pi(b_0) = \mathbb{E} \left[\sum_{i=0}^{\infty} \delta^i r_A(b_i, \pi(b_i)) \right] = \sum_{i=0}^{\infty} \delta^i \mathbb{E}[r_A(b_i, \pi(b_i))] = \sum_{i=0}^{\infty} \delta^i \phi_i^\pi(b_0) \quad (25)$$

Thus, we can rewrite (24), as

$$\sum_{i=0}^{\infty} \delta^i \phi_i^*(x_0) \leq t \left[\sum_{i=0}^{\infty} \delta^i \phi_i^{\tilde{\pi}}(m_0) \right] + (1-t) \left[\sum_{i=0}^{\infty} \delta^i \phi_i^{\tilde{\pi}}(n_0) \right] \quad (26)$$

To prove this, we show that for all $i = 0, 1, 2, \dots$

$$\phi_i^*(x_0) \leq t\phi_i^{\tilde{\pi}}(m_0) + (1-t)\phi_i^{\tilde{\pi}}(n_0) \quad (27)$$

leading directly to (26), (24), and finally the convexity of $V^*(\cdot)$ with respect to public belief.

Now let us consider general $i = k$. $\phi_{k-1}^\pi(x_0)$ will have 2^{k-1} terms each corresponding to one path of possible actions in the set $\{G, B\}^{k-2}$. Thus, at epoch $k-1$, the path λ will yield the following term in $\phi_{k-1}^*(x_0)$

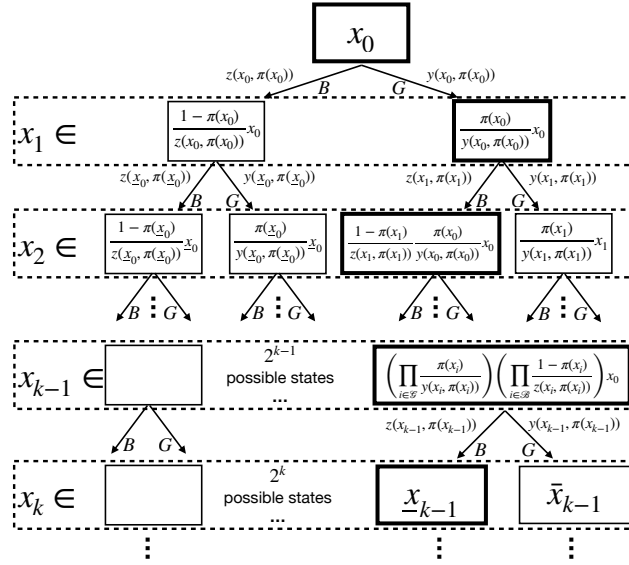
$$\left(\prod_{i \in \mathcal{G}} y(x_i, \pi_A^*(x_i)) \right) \left(\prod_{i \in \mathcal{B}} z(x_i, \pi_A^*(x_i)) \right) r_A(x_{k-1}, \pi_A^*(x_{k-1})) \quad (28)$$

We now define $P_\lambda^*(x_0) = \left(\prod_{i \in \mathcal{G}} y(x_i, \pi_A^*(x_i)) \right) \left(\prod_{i \in \mathcal{B}} z(x_i, \pi_A^*(x_i)) \right)$, that is, the probability of following the specified path λ when beginning with state x_0 and following policy π_A^* . Again, we will use $\bar{x}_i$ and $\underline{x}_i$ to represent updated beliefs from x_i after observing action G or B , respectively. Thus, $\bar{x}_i$ and $\underline{x}_i$ constitute belief values of the children nodes (in the $(i+1)$ th level) of the node with belief value x_i in the i -th level. Refer to levels $k-1$ and k in figure 3a. Considering the children $\bar{x}_{k-1}$ and $\underline{x}_{k-1}$ (in the k -th level) of the node x_{k-1} (in the $(k-1)$ -th level), we obtain the following two terms in $\phi_k^*(x_0)$:

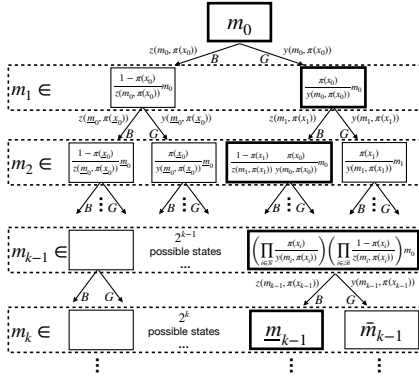
$$P_\lambda^*(x_0) \left[y(x_{k-1}, \pi_A^*(x_{k-1})) r_A(\bar{x}_{k-1}, \pi_A^*(\bar{x}_{k-1})) + z(x_{k-1}, \pi_A^*(x_{k-1})) r_A(\underline{x}_{k-1}, \pi_A^*(\underline{x}_{k-1})) \right] \quad (29)$$

Similarly, we define $P_\lambda^{\tilde{\pi}}(m_0) = \left(\prod_{i \in \mathcal{G}} y(m_i, \tilde{\pi}(m_i)) \right) \left(\prod_{i \in \mathcal{B}} z(m_i, \tilde{\pi}(m_i)) \right)$, that is, the probability of following the specified path λ when beginning with state m_0 and following policy $\tilde{\pi}$. Because $\tilde{\pi}(m_i) = \pi_A^*(x_i)$,

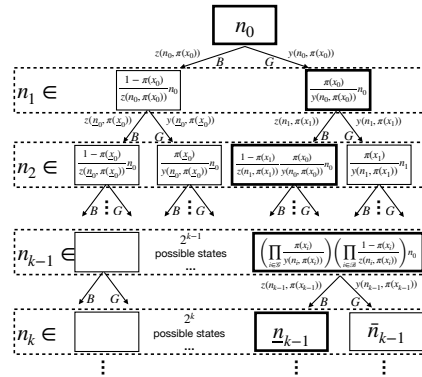
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(a) Decision tree Λ rooted at x_0



(b) Decision tree Λ_m rooted at m_0



(c) Decision tree Λ_n rooted at n_0

Figure 3: Belief evolution represented by three binary trees rooted at states x_0 , m_0 , and n_0 , respectively. Level i contains the 2^i states reachable after i epochs. Left branches correspond to action B and lead to updated beliefs $\underline{x}_0$ (respectively, $\underline{m}_0$ and $\underline{n}_0$), whereas right branches correspond to action G and lead to updated beliefs $\bar{x}_0$ (respectively, $\bar{m}_0$ and $\bar{n}_0$). Branch labels give the corresponding probabilities, and bold states mark an illustrative path such as λ . In the illustrated path, $0 \in \mathcal{G}$ and $1, k-1 \in \mathcal{B}$; bold states at each level of the three trees are counterparts.

$P_\lambda^\pi(m_0) = (\prod_{i \in \mathcal{G}} y(m_i, \pi_A^*(x_i))) (\prod_{i \in \mathcal{B}} z(m_i, \pi_A^*(x_i)))$. We also define $P_\lambda^\pi(n_0)$ by replacing $\{m_i\}$ with $\{n_i\}$.

The corresponding terms in $\phi_k^\pi(m_0)$ and $\phi_k^\pi(n_0)$ are

$$P_\lambda^*(m_0) [y(m_{k-1}, \pi_A^*(x_{k-1}))r_A(\bar{m}_{k-1}, \pi_A^*(\bar{x}_{k-1})) + z(m_{k-1}, \pi_A^*(x_{k-1}))r_A(\underline{m}_{k-1}, \pi_A^*(\underline{x}_{k-1}))] \quad (30)$$

$$P_\lambda^\pi(n_0) [y(n_{k-1}, \pi_A^*(x_{k-1}))r_A(\bar{n}_{k-1}, \pi_A^*(\bar{x}_{k-1})) + z(n_{k-1}, \pi_A^*(x_{k-1}))r_A(\underline{n}_{k-1}, \pi_A^*(\underline{x}_{k-1}))] \quad (31)$$

We will show that

$$\begin{aligned} & P_\lambda^*(x_0) [y(x_{k-1}, \pi_A^*(x_{k-1}))r_A(\bar{x}_{k-1}, \pi_A^*(\bar{x}_{k-1})) + z(x_{k-1}, \pi_A^*(x_{k-1}))r_A(\underline{x}_{k-1}, \pi_A^*(\underline{x}_{k-1}))] \\ & \leq tP_\lambda^*(m_0) [y(m_{k-1}, \pi_A^*(x_{k-1}))r_A(\bar{m}_{k-1}, \pi_A^*(\bar{x}_{k-1})) + z(m_{k-1}, \pi_A^*(x_{k-1}))r_A(\underline{m}_{k-1}, \pi_A^*(\underline{x}_{k-1}))] \\ & + (1-t)P_\lambda^\pi(n_0) [y(n_{k-1}, \pi_A^*(x_{k-1}))r_A(\bar{n}_{k-1}, \pi_A^*(\bar{x}_{k-1})) + z(n_{k-1}, \pi_A^*(x_{k-1}))r_A(\underline{n}_{k-1}, \pi_A^*(\underline{x}_{k-1}))] \end{aligned} \quad (32)$$

Similarly, each pair of children for each term in the $(k-1)$ -th level of the belief tree can be combined and the corresponding inequalities can be obtained following the same process. Summing the resulting inequalities

will yield (27), leading to (26), (24), and the convexity of $V^*(\cdot)$.

Proof of (32)

We now focus on the latter of the two terms of (29). We will show the following two properties

$$\begin{aligned} \underline{x}_{k-1} = & t \left(\prod_{i \in \mathcal{G}} \frac{y(m_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(m_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right) \underline{m}_{k-1} \\ & + (1-t) \left(\prod_{i \in \mathcal{G}} \frac{y(n_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(n_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right) \underline{n}_{k-1} \end{aligned} \quad (33)$$

$$\begin{aligned} 1 = & t \left(\prod_{i \in \mathcal{G}} \frac{y(m_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(m_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right) \\ & + (1-t) \left(\prod_{i \in \mathcal{G}} \frac{y(n_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(n_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right) \end{aligned} \quad (34)$$

Defining $\tilde{t} = t \left(\prod_{i \in \mathcal{G}} \frac{y(m_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(m_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right)$, it follows from (33) and (34) that

$$\underline{x}_{k-1} = \tilde{t} \underline{m}_{k-1} + (1-\tilde{t}) \underline{n}_{k-1} \quad (35)$$

From the expressions for $\tilde{t}$, $P_\lambda^*(x_0)$, $P_\lambda^{\tilde{\pi}}(m_0)$, it follows that:

$$\tilde{t} z(x_{k-1}, \pi_A^*(x_{k-1})) P_\lambda^*(x_0) = t z(m_{k-1}, \pi_A^*(x_{k-1})) P_\lambda^{\tilde{\pi}}(m_0) \quad (36)$$

Similarly, we can show

$$\tilde{t} z(x_{k-1}, \pi_A^*(x_{k-1})) P_\lambda^*(x_0) = (1-\tilde{t}) z(n_{k-1}, \pi_A^*(x_{k-1})) P_\lambda^{\tilde{\pi}}(n_0) \quad (37)$$

We apply this as follows to the latter term of (29):

$$\begin{aligned} & P_\lambda^*(x_0) z(x_{k-1}, \pi_A^*(x_{k-1})) r_A(\underline{x}_{k-1}, \pi_A^*(\underline{x}_{k-1})) \\ & \leq P_\lambda^*(x_0) z(x_{k-1}, \pi_A^*(x_{k-1})) \\ & \quad [\tilde{t} r_A(\underline{m}_{k-1}, \pi_A^*(\underline{x}_{k-1})) + (1-\tilde{t}) r_A(\underline{n}_{k-1}, \pi_A^*(\underline{x}_{k-1}))] \quad (\text{Convexity of } r, \text{Eqn 35}) \\ & = t \left[P_\lambda^{\tilde{\pi}}(m_0) z(m_{k-1}, \pi_A^*(x_{k-1})) r_A(\underline{m}_{k-1}, \pi_A^*(\underline{x}_{k-1})) \right] \\ & \quad + (1-t) \left[P_\lambda^{\tilde{\pi}}(n_0) z(n_{k-1}, \pi_A^*(x_{k-1})) r_A(\underline{n}_{k-1}, \pi_A^*(\underline{x}_{k-1})) \right] \quad (\text{Eqns 36, 37}) \end{aligned} \quad (38)$$

Applying the same argument to the first term of (29) yields:

$$\begin{aligned} & P_\lambda^*(x_0) y(x_{k-1}, \pi_A^*(x_{k-1})) r_A(\bar{x}_{k-1}, \pi_A^*(\bar{x}_{k-1})) \\ & \leq t \left[P_\lambda^{\tilde{\pi}}(m_0) y(m_{k-1}, \pi_A^*(x_{k-1})) r_A(\bar{m}_{k-1}, \pi_A^*(\bar{x}_{k-1})) \right] \\ & \quad + (1-t) \left[P_\lambda^{\tilde{\pi}}(n_0) y(n_{k-1}, \pi_A^*(x_{k-1})) r_A(\bar{n}_{k-1}, \pi_A^*(\bar{x}_{k-1})) \right] \end{aligned} \quad (39)$$

Summing (38) and (39) yields (32).

We now complete the proof of (32) and, therefore, the entire proof, by proving (33) and (34).

Proof of (33)

From the recurrence relation in Equation (13) and Equation (3), we can write $\underline{x}_{k-1}$, $\underline{m}_{k-1}$, and $\underline{n}_{k-1}$ as the product of all belief updates with x_0 , m_0 , and n_0 , respectively, as follows. Each term in the products corresponds to the multiplicative state update after a single action.

$$\underline{x}_{k-1} = \left(\prod_{i \in \mathcal{G}} \frac{\pi_A^*(x_i)}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{1 - \pi_A^*(x_i)}{z(x_i, \pi_A^*(x_i))} \right) x_0 \quad (40)$$

$$\underline{m}_{k-1} = \left(\prod_{i \in \mathcal{G}} \frac{\pi_A^*(x_i)}{y(m_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{1 - \pi_A^*(x_i)}{z(m_i, \pi_A^*(x_i))} \right) m_0 \quad (41)$$

$$\underline{n}_{k-1} = \left(\prod_{i \in \mathcal{G}} \frac{\pi_A^*(x_i)}{y(n_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{1 - \pi_A^*(x_i)}{z(n_i, \pi_A^*(x_i))} \right) n_0 \quad (42)$$

These equations may be best understood via the illustration in figure 3. We then reorganize terms to relate $\underline{x}_{k-1}$ to $\underline{m}_{k-1}$ and $\underline{n}_{k-1}$, using the fact that $x_0 = tm_0 + (1-t)n_0$.

$$\begin{aligned}\underline{x}_{k-1} &= \left(\prod_{i \in \mathcal{G}} \frac{\pi_A^*(x_i)}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{1 - \pi_A^*(x_i)}{z(x_i, \pi_A^*(x_i))} \right) [tm_0 + (1-t)n_0] \quad (\text{from 40}) \\ &= t \left(\prod_{i \in \mathcal{G}} \frac{y(m_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(m_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right) \underline{m}_{k-1} \\ &\quad + (1-t) \left(\prod_{i \in \mathcal{G}} \frac{y(n_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(n_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right) \underline{n}_{k-1} \quad (\text{from 41 and 42}) \quad (43)\end{aligned}$$

This proves (33).

Proof of (34)

We first consider the numerator of the first term in (34): $\prod_{i \in \mathcal{G}} y(m_i, \pi_A^*(x_i)) \prod_{i \in \mathcal{B} \cup \{k-1\}} z(m_i, \pi_A^*(x_i))$. We now prove that this can be written as $\alpha_0 m_0 + \beta_0$ for some α_0, β_0 which depend only on $\{\pi_A^*(x_i)\}_{i=1, \dots, k-1}$ and the nature of the dependence is determined by the sequence of actions along this path in stages 1 to $k-1$. We will prove this via induction working backwards from the final term (i.e. the $(k-1)$ -th term) of the product. Therefore our base case will be the final term of this product which can be either $y(m_{k-1}, \pi_A^*(x_{k-1}))$ or $z(m_{k-1}, \pi_A^*(x_{k-1}))$, depending on the preceding action (in the $(k-1)$ -th stage).

$$\begin{aligned}y(m_{k-1}, \pi_A^*(x_{k-1})) &= 1 + 2m_{k-1}\pi_A^*(x_{k-1}) - m_{k-1} - \pi_A^*(x_{k-1}) \\ &= (2\pi_A^*(x_{k-1}) - 1)m_{k-1} + (1 - \pi_A^*(x_{k-1})) \\ z(m_{k-1}, \pi_A^*(x_{k-1})) &= m_{k-1} + \pi_A^*(x_{k-1}) - 2m_{k-1}\pi_A^*(x_{k-1}) \\ &= (1 - 2\pi_A^*(x_{k-1}))m_{k-1} + (\pi_A^*(x_{k-1}))\end{aligned}$$

Thus, the final term of the product is of the form $\alpha_{k-1}m_{k-1} + \beta_{k-1}$ where α_{k-1} and β_{k-1} depend only upon $\pi_A^*(x_{k-1})$, and the nature of the dependence is determined entirely by the $(k-1)$ -th action (since this action determines whether the final term is $y(\cdot)$ or $z(\cdot)$).

For the inductive step, assume that $\prod_{i \in \mathcal{G}, i > j} y(m_i, \pi_A^*(x_i)) \prod_{i \in \mathcal{B} \cup \{k-1\}, i > j} z(m_i, \pi_A^*(x_i))$ can be written as $\alpha_{j+1}m_{j+1} + \beta_{j+1}$ for some $\alpha_{j+1}, \beta_{j+1}$ which depend only on $\{\pi_A^*(x_i)\}_{i > j}$ and the nature of the dependence is determined entirely by the sequence of actions in this path for $i > j$ to $i = k-1$. We will now show that $\prod_{i \in \mathcal{G}, i > j-1} y(m_i, \pi_A^*(x_i)) \prod_{i \in \mathcal{B} \cup \{k-1\}, i > j-1} z(m_i, \pi_A^*(x_i))$ can be written as $\alpha_j m_j + \beta_j$ again with α_j, β_j depending only on $\{\pi_A^*(x_i)\}_{i > j-1}$ with the nature of the dependence determined by the sequence of actions on

this path for $i > j-1$ to $i = k-1$. We define $f(b, q, a) = \begin{cases} y(b, q) & a = G \\ z(b, q) & a = B \end{cases}$ and $g(q, a) = \begin{cases} q & a = G \\ 1 - q & a = B \end{cases}$.

Thus,

$$\begin{aligned}&\prod_{i \in \mathcal{G}, i > j-1} y(m_i, \pi_A^*(x_i)) \prod_{i \in \mathcal{B} \cup \{k-1\}, i > j-1} z(m_i, \pi_A^*(x_i)) \\ &= f(m_j, \pi_A^*(x_j), a_{j+1}) \prod_{i \in \mathcal{G}, i > j} y(m_i, \pi_A^*(x_i)) \prod_{i \in \mathcal{B} \cup \{k-1\}, i > j} z(m_i, \pi_A^*(x_i)) \\ &= f(m_j, \pi_A^*(x_j), a_{j+1}) [\alpha_{j+1}m_{j+1} + \beta_{j+1}] \quad (\text{inductive hypothesis}) \\ &= f(m_j, \pi_A^*(x_j), a_{j+1}) \left[\alpha_{j+1} \frac{g(\pi_A^*(x_j), a_{j+1})}{f(m_j, \pi_A^*(x_j), a_{j+1})} m_j + \beta_{j+1} \right] \quad (\text{Equation (13)}) \\ &= \alpha_{j+1}g(\pi_A^*(x_j), a_{j+1})m_j + \beta_{j+1}f(m_j, \pi_A^*(x_j), a_{j+1})\end{aligned}$$

Thus, from the definitions of $f(\cdot), g(\cdot), y(\cdot)$, and $z(\cdot)$,

$$\alpha_j = \begin{cases} \alpha_{j+1}\pi_A^*(x_j) + \beta_{j+1}(2\pi_A^*(x_j) - 1) & a_{j+1} = G \\ \alpha_{j+1}(1 - \pi_A^*(x_j)) + \beta_{j+1}(1 - 2\pi_A^*(x_j)) & a_{j+1} = B \end{cases} \text{ and } \beta_j = \begin{cases} \beta_{j+1}(1 - \pi_A^*(x_j)) & a_{j+1} = G \\ \beta_{j+1}\pi_A^*(x_j) & a_{j+1} = B \end{cases}$$

In both cases, by our inductive hypothesis, $\alpha_{j+1}, \beta_{j+1}$ depend only on $\{\pi_A^*(x_i)\}_{i > j}$ and the actions in this path from $i > j$ to $i = k-1$. Thus, the claim holds for α_j, β_j , completing the induction.

Now recall that the denominator of (34) is $\prod_{i \in \mathcal{G}} y(x_i, \pi_A^*(x_i)) \prod_{i \in \mathcal{B} \cup \{k-1\}} z(x_i, \pi_A^*(x_i))$. Note that the policy and sequence of actions is the same as those for the numerator. Thus, because, as shown, the coefficients α_0, β_0 depend only on $\{\pi_A^*(x_i)\}$ and the path of actions, the same logic can be applied to write the denominator

as $\alpha_0 x_0 + \beta_0$ for the same α_0, β_0 . This can be done similarly for the numerator of the second term in (34). Thus, we can reason about the coefficients in (33) as follows:

$$\begin{aligned}
& t \left(\prod_{i \in \mathcal{G}} \frac{y(m_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(m_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right) \\
& + (1-t) \left(\prod_{i \in \mathcal{G}} \frac{y(n_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(n_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right) \\
& = \frac{t(\alpha_0 m_0 + \beta_0) + (1-t)(\alpha_0 n_0 + \beta_0)}{\alpha_0 x_0 + \beta_0} \\
& = \frac{\alpha_0 x_0 + \beta_0}{\alpha_0 x_0 + \beta_0} \quad (\text{Since } x_0 = t m_0 + (1-t) n_0) \\
& = 1
\end{aligned}$$

Thus, (34) holds, completing the proof. $\square$

C.4 ALTRUISTIC MYOPIC POLICY BOUND

Lemma 9.

$$\pi_A^0(b) \leq \pi_A^*(b) \quad \forall b \in [0, 1]$$

Assume by way of contradiction that there exists $b \in [0, 1]$ such that $\pi_A^0(b) > \pi_A^*(b)$. We will show that $V_A^*(b) < V_A^\pi(b)$ for a policy π we construct, violating the optimality of $V_A^*(\cdot)$.

Let π be such that, when starting at public belief b , the policy π applies precision $\pi_A^0(b)$ at the current time step and then applies the optimal policy at all future time steps. We now elaborate $V_A^*(b) - V_A^\pi(b)$ by breaking the value function into instantaneous reward r_A and future discounted reward. We will refer to b^+ and b^- as the updated beliefs after receiving signals G and B , respectively when applying the optimal policy and b_m^+ and b_m^- as the same when applying the myopic policy.

$$\begin{aligned}
V_A^*(b) - V_A^\pi(b) &= r_A(b, \pi_A^*(b)) - r_A(b, \pi_A^0(b)) \\
&+ \delta y(b, \pi_A^*(b)) V_A^*(b^+) + \delta z(b, \pi_A^*(b)) V_A^*(b^-) \\
&- \delta y(b, \pi_A^0(b)) V_A^*(b_m^+) - \delta z(b, \pi_A^0(b)) V_A^*(b_m^-) \\
&< \delta y(b, \pi_A^*(b)) V_A^*(b^+) + \delta z(b, \pi_A^*(b)) V_A^*(b^-) \\
&- \delta y(b, \pi_A^0(b)) V_A^*(b_m^+) - \delta z(b, \pi_A^0(b)) V_A^*(b_m^-) \quad (\text{definition of } \pi_A^0) \\
&\leq 0 \quad (\text{Theorem 2})
\end{aligned}$$

Here, we applied the fact that the instantaneous reward under $\pi_A^0(\cdot)$ must be greater from the definition of the myopic optimal policy.

We then relied on the convexity of $V_A^*(\cdot)$ from Theorem 2 for the future cost. From Equation (13), note that $b_m^- \leq b^- \leq b \leq b^+ \leq b_m^+$. Along with this convexity, this leads to the last step above and completes the proof. $\blacksquare$

C.5 PROOF OF THEOREM 3: OPTIMAL ALTRUISTIC POLICY

Consider $V_A^\pi(b)$ with policy π applying precision q in the current time step and applying the optimal policy at all future time steps. The q that maximizes $V_A^\pi(b)$ provides the optimum precision starting at public belief b . Because the planner never benefits from decreasing precision below the baseline p (see Appendix C.9), we restrict q to $[p, 1]$ without loss of generality.

We will let b^+ and b^- denote the positive and negative belief updates possible after receiving a signal with precision q . Taking the second derivative of $V_A^\pi(b)$ with respect to q for $q \geq \max(b, 1-b)$:

$$\frac{\partial^2}{\partial q^2} V_A^\pi(b) = \begin{cases} -\ddot{\beta}(q) + \delta b^2 (1-b)^2 \left[\frac{\ddot{V}_A^*(b^+)}{y^3(b, q)} + \frac{\ddot{V}_A^*(b^-)}{z^3(b, q)} \right] & q \geq \max(b, 1-b) \\ -\ddot{\beta}(q) & \text{otherwise} \end{cases}$$

With the convexity and concavity of $V_A^*(\cdot)$ and $\beta(\cdot)$, respectively, the above shows that $V_A^\pi(b)$ is convex with respect to q on $[\max(b, 1-b), 1]$. Thus, on this interval, it is maximized at one of the two extreme points.

For $q \in [0.5, \max(b, 1 - b))$, the derivative of $V_A^\pi(b)$ with respect to q is negative; thus, the optimal choice will be the baseline precision p to minimize the cost. Thus, there are three possible optimal precisions: p , $\max(b, 1 - b)$, and 1.

Note that the expected value under the latter two candidates is always strictly negative. Under precision p , however, the expected value is 0 at $b = 0$ and $b = 1$. This, combined with the non-positivity and symmetry of the value function (Lemma 8) implies the existence of d_A such that $\pi_A^*(b) = p$ for $b \in [0, d_A] \cup (1 - d_A, 1]$. Furthermore, from Theorem 1 and applying Lemma 9, $d_A \in (0, t_M)$ where t_M is defined as in the statement of Theorem 1.

Finally, Theorem 1 and applying Lemma 9, also imply the existence of $t_A \in [d_A, t_M]$ such that $\pi_A^*(b) = 1$ for $b \in (t_A, 1 - t_A)$. This completes the proof. ■

C.6 PROOF OF THEOREM 4: MYOPIC BIASED POLICY

Preliminaries Applying the action rule from (2), the instantaneous reward (6) can be written as follows:

$$r_B(b, q) = \begin{cases} -\beta(|q - p|) - Cz(b, q) & q \geq \max(b, 1 - b) \\ -\beta(|q - p|) - C & b < 1 - q \\ -\beta(|q - p|) & b > q \end{cases} \quad (44)$$

Note that $z(b, q) = b + q - 2bq$ increases with respect to q when $b < 0.5$ and decreases with respect to q when $b > 0.5$.

At $b = 0.5$, $z(b, q) = z(0.5, q) = 0.5$ regardless of the precision chosen. Furthermore, at this public belief, $q \geq \max(b, 1 - b)$ because $q \geq 0.5$ for any q , falling in the first case of (44). In this case, the second term of $r_B(b, q)$ is unchanging with respect to q , so the planner seeks to maximize only the first term, which is accomplished at $q = p$.

We now prove a series of lemmas (10-14) which will assist us in our proof of Theorem 4.

Lemma 10.

If $\pi_B^0(b) \geq \max(b, 1 - b)$, then $\pi_B^0(b) \in \{\max(b, 1 - b), p, 1\}$.

Proof. The reward is differentiable everywhere except $q = p$. Thus, we can write its derivative with respect to the chosen precision as follows (except at $q = p$):

$$\frac{\partial r_B(b, q)}{\partial q} = \begin{cases} -\text{sign}(q - p)\dot{\beta}(|q - p|) & q \geq \max(b, 1 - b) \\ -C(1 - 2b) & \\ -\text{sign}(q - p)\dot{\beta}(|q - p|) & \text{otherwise} \end{cases} \quad (45)$$

In the former case, we can write the second derivative of the reward with respect to q as follows, for $q \neq p$:

$$\frac{\partial^2}{\partial q^2} r_B(b, q) = -\ddot{\beta}(|q - p|)$$

Because $\beta(\cdot)$ is concave, this expression is non-negative, and $r_B(b, q)$ is convex with respect to q . Thus, the reward is maximized at one of the extreme points $\{\max(b, 1 - b), 1\}$ or p because $r_B(b, q)$ is not differentiable at $q = p$. □

Lemma 11.

For $b \in [0, 1 - p)$, $\pi_B^0(b) \in \{p, 1 - b\}$.

Proof. Because $p > 0.5$, for $b \in [0, 1 - p)$, $\max(b, 1 - b) = 1 - b > p$.

First, consider the case when $\pi_B^0(b) < \max(b, 1 - b) = 1 - b$. From (44), in such instances, the reward is maximized by minimizing $|q - p|$. Thus, the optimal precision lower than $\max(b, 1 - b)$ is p .

Now consider $\pi_B^0(b) \geq \max(b, 1 - b) = 1 - b$. By Lemma 10, $\pi_B^0(b) \in \{1 - b, p, 1\}$. What remains is to rule out $\pi_B^0(b) = 1$. To do so, we compare $r_B(b, 1)$ and $r_B(b, 1 - b)$. Both fall under the first case of (44). Because $\beta(\cdot)$ is increasing in its argument and $p < 1 - b \leq 1$, the first term of the reward from (44) is greater for precision $1 - b$ than for precision 1. Since $z(b, q)$ is increasing with respect to q for $b < 0.5$, $r_B(b, 1) \leq r_B(b, \max(b, 1 - b))$ so we can rule out precision 1. This completes the proof. □

Lemma 12.

For $b \in (p, 1]$, $\pi_B^0(b) = p$.

Proof. The reward is strictly negative for any $q \neq p$ because of the term $-\beta(|q - p|)$. For $b \in (p, 1]$ and $q = p$, $p < \max(b, 1 - b) = b$, leading to the third case of (44) and yielding reward 0. Thus, the optimal precision is p . $\square$

Lemma 13.

For $b \in [1 - p, 0.5]$, $\pi_B^0(b) \in \{p, 1 - b\}$.

Proof. When $p = 0.5$, the interval $[1 - p, 0.5]$ contains only $\{0.5\}$. At $b = 0.5$ $p = 1 - b$, thus $\pi_B^0(b) = p = 1 - b = 0.5$. We deal with $p > 0.5 \geq b$ for the remainder of the proof.

For $b \in [1 - p, 0.5]$, $\max(b, 1 - b) = 1 - b \leq p$. Thus, from the first case of (44), precision p yields reward $r_B(b, p) = -Cz(b, p)$. We now consider two cases: $q < \max(b, 1 - b)$ and $q \geq \max(b, 1 - b)$.

Suppose $q < \max(b, 1 - b) = 1 - b$. This leads to the second case of (44) and yields reward $-\beta(|q - p|) - C$. Furthermore, $z(b, q)$ is increasing with respect to q for $b < 0.5$ and takes values between $z(b, 0.5) = 0.5$ and $z(b, 1) = 1 - b$. Thus, $z(b, q) < 1$ for all q . Therefore, the reward from policy $q < 1 - b$ is strictly lower than $r_B(b, p)$, and we can rule out this case.

When $q \geq \max(b, 1 - b)$, applying Lemma 10 yields that the remaining candidate precisions are $\{1 - b, p, 1\}$. We rule out precision 1 by comparing it to precision p . The first term of the reward from (44) is greater for precision p than for precision 1. Since $z(b, q)$ is increasing with respect to q for $b < 0.5$, the second term is also greater for precision p , and we can rule out precision 1. This completes the proof. $\square$

We now define the notion of ϵ -optimal policies.

Definition 1. ϵ -optimal policies

An ϵ -optimal policy $\pi^\epsilon(b)$ is any policy such that the following holds

$$V^*(b) \leq V^{\pi^\epsilon}(b) + \epsilon$$

for $\epsilon > 0$.

Note that from the definition, any optimal policy is also ϵ -optimal for any $\epsilon > 0$. Thus, $\pi_B^0(b)$ is also ϵ -optimal

Following this definition, let $\pi_B^{\epsilon, 0}(b)$ denote an ϵ -optimal myopic policy for the biased planner.

Lemma 14.

For $b \in (0.5, p]$ and any $\epsilon > 0$, there are two possibilities:

1. $\pi_B^0(b) \in \{p, 1\}$, or

2. $\pi_B^{\epsilon, 0}(b) = b - \epsilon$

Proof. For $b \in (0.5, p]$, $\max(b, 1 - b) = b \leq p$. We consider two cases: $q \geq \max(b, 1 - b)$ or $q < \max(b, 1 - b)$.

In the first case, Lemma 10 yields that $\pi_B^0(b) \in \{b, p, 1\}$. We rule out precision b by comparing it to precision p . The first terms of the reward from (44) is greater for precision p . Furthermore, since $z(b, q)$ is decreasing with respect to q for $b > 0.5$, the second term of the reward is also greater for precision p . Thus, we can rule out precision b . This yields the first claim.

In the latter case, $q < \max(b, 1 - b) = b$. Therefore, (1) since $q \geq 0.5$, $q \in [0.5, b)$, and (2) the third case of (44) applies leading to reward $-\beta(|q - p|)$. Thus, the reward $r_B(b, q)$ is an increasing function of $q \in [0.5, b)$ because $0.5 < b \leq p$. This leads directly to the second claim. $\square$

Remark 3. As noted in Puterman (1990) Section 2.3.1, an optimal policy may not exist for infinite action spaces because the supremum may not be attained. When an optimal policy does not exist, we instead seek ϵ -optimal policies, i.e., policies that yield reward within $\epsilon > 0$ of the supremum. In particular, this is true for case 2 of Lemma 14.

Proof of Theorem 4 We consider the myopic biased policy in 4 cases: $b \in [0, 1 - p]$, $b \in [1 - p, 0.5]$, $b \in (0.5, p]$, and $b \in (p, 1]$, making use of the fact that $p \geq 0.5$. Based on Lemmas 11-14, these cases have the following candidate policies: $\{p, 1 - b\}$, $\{1 - b, p\}$, $\{1, p, b - \epsilon\}$, and $\{p\}$, respectively.

Case 1: $b \in [0, 1 - p]$, $\pi_B^0(b) \in \{p, 1 - b\}$ (Lemma 11)

We will show the existence of a threshold t_1 such that $\pi_B^0(b) = p$ for $b \in [0, t_1]$ and $\pi_B^0(b) = 1 - b$ for $b \in (t_1, 1 - p)$.

We can express the reward under the two candidate policies as follows by applying (44):

$$\begin{aligned} r_B(b, p) &= -C \\ r_B(b, 1 - b) &= -\beta(1 - b - p) - C(1 - 2b(1 - b)) \end{aligned}$$

The derivative of $r_B(b, 1 - b)$ with respect to b is

$$\frac{\partial}{\partial b} r_B(b, 1 - b) = \dot{\beta}(1 - p - b) + 2C(1 - 2b)$$

which is positive as $\beta(\cdot)$ is increasing and $b < 1 - p \leq 0.5$. Furthermore, when evaluated at $b = 0$ and $b = 1 - p$ we obtain

$$\begin{aligned} r_B(0, 1) &= -\beta(1 - p) - C < r_B(b, p) \\ r_B(1 - p, p) &= -C(1 - 2p(1 - p)) > r_B(b, p) \end{aligned}$$

Thus, $r_B(b, 1 - b)$ is increasing on $[0, 1 - p]$ with $r_B(b, 1 - b) < r_B(b, p)$ for $b = 0$ and vice versa for $b = 1 - p$. Because $\beta(\cdot)$ is continuous, so are both reward functions. Thus, there exists $t_1 \in (0, 1 - p)$ such that $\pi_B^0(b) = p$ for $b \in [0, t_1]$ and $\pi_B^0(b) = 1 - b$ for $b \in (t_1, 1 - p)$.

Case 2: $b \in [1 - p, 0.5]$, $\pi_B^0(b) \in \{1 - b, p\}$ (Lemma 13)

If $p = 0.5$, this interval contains only $\{0.5\}$, and, for $b = 0.5$, $1 - b = p$. Thus, the two candidate policies are equivalent, and the optimal policy is $\pi_B^0(b) = p$. For the remainder of this case, we consider $p > 0.5$.

We now prove the existence of t_2, t_3 such that $\pi_B^0(b) = 1 - b$ for $b \in [t_2, t_3]$, and $\pi_B^0(b) = p$ for $b \in [1 - p, t_2] \cup [t_3, 0.5]$.

We can express the reward under the candidate policies as follows by applying (44):

$$\begin{aligned} r_B(b, 1 - b) &= -\beta(p - 1 + b) - Cz(b, 1 - b) \\ r_B(b, p) &= -Cz(b, p) \end{aligned}$$

This leads to the following condition for the optimal policy to be $1 - b$:

$$\begin{aligned} r_B(b, 1 - b) &\geq r_B(b, p) \\ \beta(p - 1 + b) &\leq C[z(b, p) - z(b, 1 - b)] \\ \beta(p - 1 + b) &\leq C(p - 1 + b)(1 - 2b) \end{aligned} \tag{46}$$

The LHS of (46) is a concave, increasing function of b on $[1 - p, 0.5]$. The RHS of (46) is a concave parabola that increases in the first half of $[1 - p, 0.5]$ and decreases in the second half.

Now consider the two halves of the interval $[1 - p, 0.75 - 0.5p]$ and $[0.75 - 0.5p, 0.5]$. In the first half, the LHS and RHS of (46) are increasing and concave, with the RHS also being a parabola. Thus, in the first half, the two intersect at most two points (one of which is at $b = 1 - p$). In the second half, the LHS of (46) is increasing while the RHS is decreasing. Thus, they intersect at most once. Let the two possible intersections for $b > 1 - p$ be t_2 and t_3 with $t_2 \leq t_3$. Thus, what remains is to determine the optimal on each of $(1 - p, t_2)$, (t_2, t_3) , and $(t_3, 0.5)$. At $b = 0.5$, $r_B(b, p) > r_B(b, 1 - b)$ because $p > 0.5$, thus $\pi_B^0(b) = p$ for $b \in (t_3, 0.5)$. Subsequently, $\pi_B^0(b) = 1 - b$ for (t_2, t_3) , and $\pi_B^0(b) = p$ for $(1 - p, t_2)$.

Case 3: $b \in (0.5, p]$, $\pi_B^{\epsilon, 0}(b) \in \{1, p, b - \epsilon\}$ (Lemma 14)

In this case, we show the existence of $0.5 \leq t_4 \leq t_5 \leq p$ such that the optimal precisions on intervals $(0.5, t_4]$, (t_4, t_5) , and $[t_5, p)$ are p , 1 , and $b - \epsilon$, respectively.

We can express the reward under the candidate policies as follows by applying (44):

$$\begin{aligned} r_B(b, 1) &= -\beta(1 - p) - C(1 - b) \\ r_B(b, p) &= -Cz(b, p) \\ r_B(b, b - \epsilon) &= -\beta(p - b + \epsilon) \end{aligned}$$

When evaluated at $b = p$, $r_B(b, b - \epsilon)$ is arbitrarily close to 0, and, therefore, $b - \epsilon$ is the optimal precision on some interval $(t_5, p]$ for $0.5 \leq t_5 < p$.

Now, note that $r_B(b, 1)$ and $r_B(b, p)$ are linear functions in b while $r_B(b, b - \epsilon)$ is convex in b (from the concavity of β). Thus, $r_B(b, b - \epsilon) - r_B(b, p)$ and $r_B(b, b - \epsilon) - r_B(b, 1)$ are convex and, as a result, have convex level sets. This implies that the set on which both of these expressions are positive, i.e., the set for which $b - \epsilon$ is the optimal policy, must also be convex. Therefore, $b - \epsilon$ cannot be optimal on any subset of $(0.5, t_5)$.

What remains is to compare precisions 1 and p on the interval $(0.5, t_5]$. To do so, we inspect the values of $r_B(b, 1)$ and $r_B(b, p)$ at 0.5 and their derivatives.

$$\begin{aligned} r_B(0.5, 1) &= -\beta(1 - p) - 0.5C \\ r_B(0.5, p) &= -0.5C \end{aligned}$$

Because $r_B(0.5, 1) < r_B(0.5, p)$ there exists interval $(0.5, t_4)$ with $t_4 \leq t_5$ such that $\pi_B^0(b) = p$. Furthermore, because both are linear in b , there is at most one crossover point where their ordering changes. Thus, if $t_4 < t_5$, then $\pi_B^0(b) = 1$ on (t_4, t_5) .

Case 4: $b \in (p, 1]$, $\pi_B^0(b) = p$

This follows directly from Lemma 12 and completes the proof of Theorem 4. ■

C.7 BIASED VALUE FUNCTION MONOTONICITY

Lemma 15. For any $b_1, b_2 \in [0, 1]$ such that $b_1 \leq b_2$,

$$V_B^*(b_1) \leq V_B^*(b_2)$$

Proof. Applying (44), the derivative of the biased reward with respect to b can be written as follows:

$$\frac{\partial}{\partial b} r_B(b, q) = \begin{cases} -C(1 - 2q) & q \geq \max(b, 1 - b) \\ 0 & \text{otherwise} \end{cases} \quad (47)$$

Because $q \geq 0.5$ and $z(b, q) \in [0, 1]$, it follows that $r_B(b, q)$ is increasing with respect to b .

Next, note that the Bayesian nature of belief updates implies that the belief process is a martingale. That is, for any fixed q_i , $\mathbb{E}[b_{i+1}] = b_i$. To demonstrate this consider two cases: (1) $q_i < \max(b_i, 1 - b_i)$, and (2) $q_i \geq \max(b_i, 1 - b_i)$.

In the former case, it follows from (3) that $b_{i+1} = b_i$, satisfying the claim.

In the latter case, (13) gives the distribution of b_{i+1} which yields the same.

The monotonicity of $V_B^*(\cdot)$ then follows from Proposition 5 of Smith & McCardle (2002). □

C.8 PROOF OF THEOREM 5: OPTIMAL BIASED POLICY

We first prove that the optimal value function is continuous in the following lemma:

Lemma 16. Biased value function continuity

$V_B^*(\cdot)$ is continuous w.r.t. b .

Proof. First consider the one-stage value function defined as $V_B^0(b) = \sup_{q \in [0.5, 1]} r_B(b, q)$. From Theorem 4, we know that $\pi_B^{\epsilon, 0}(b) \in \{1 - b, 1, b - \epsilon, p\}$. Thus, we can equivalently write the one-stage value function as follows:

$$\begin{aligned} V_B^0(b) &= \sup_{\epsilon > 0} r_B(b, \pi_B^{\epsilon, 0}(b)) \\ &= \sup_{\epsilon > 0} \max_{q \in \{p, 1 - b, 1, b - \epsilon, p\}} r_B(b, q) \end{aligned} \quad (48)$$

Using the above expression, we now argue that $V_B^0(\cdot)$ is continuous on $b \in [0, 1]$. First, consider $b \leq 0.5$. From Theorem 4, an ϵ -optimal policy applies precision $1 - b$ or p for $b \leq 0.5$. In either case, the instantaneous reward will not depend upon ϵ , and the supremum in (48) can be omitted, leaving $V_B^0(b) = \max_{q \in \{1 - b, p\}} r_B(b, q)$. For $b \leq 0.5$, $r_B(b, 1 - b)$ falls in case one of (44) and is thus continuous with respect to b . On the same interval, $r_B(b, p)$ has one discontinuity at $b = 1 - p$ where it switches from case two to case one of (44). At $b = 1 - p$, Theorem 4 shows that the optimal myopic policy switches from $1 - b$ to p . Both policies fall in the first case of

(44), and, at $b = 1 - p$, $p = 1 - b$, so the controls are equal. Thus, the left and right limits coincide with the optimal value, and $V_B^0(\cdot)$ is continuous.

For $b \geq 0.5$, Theorem 4 gives the candidate ϵ -optimal policies $q \in \{1, b - \epsilon, p\}$. When the inner maximum in (48) is attained for $q = b - \epsilon$, the resulting value is $V_B^0(b) = \sup_{\epsilon > 0} r_B(b, b - \epsilon)$. From the third case of (44), we can rewrite this as $V_B^0(b) = \sup_{\epsilon > 0} -\beta(|b - \epsilon - p|)$. Since $b - \epsilon$ is only selected for $b \leq p$ (Theorem 4), this reduces to $V_B^0(b) = -\beta(|p - b|)$. Thus, exchanging the order of the maximum and supremum in (48), we obtain $V_B^0(b) = \max\{r_B(b, 1), -\beta(|p - b|), r_B(b, p)\}$ for $b \geq 0.5$. For the first expression in this maximum, $r_B(b, 1)$ falls into case one of (44) and is thus continuous. The continuity of $-\beta(|p - b|)$ follows from the continuity of $\beta(\cdot)$. For $r_B(b, p)$, since $b \geq 0.5$, the first case of (44) applies for $b \leq p$ and the third case applies for $b > p$. Thus, $r_B(b, p)$ is potentially discontinuous at $b = p$. Therefore, what remains is to show the continuity of $V_B^0(\cdot)$ at $b = p$.

At $b = p$, the myopic ϵ -optimal policy switches from $q = b - \epsilon$ to $q = p$ by Theorem 4 with $\pi_B^{\epsilon, 0}(p) = b - \epsilon$. Thus, for $b = p$, the inner maximum of (48) is attained at $q = b - \epsilon$ and, from the third case of (44), $V_B^0(p) = \sup_{\epsilon > 0} -\beta(\epsilon) = 0$. Similarly, from Theorem 4, for each $\epsilon > 0$, there exists $t_\epsilon < p$ such that on $[t_\epsilon, p]$ the inner maximum of (48) is attained by $q = b - \epsilon$. On this interval, $V_B^0(b) = \sup_{\epsilon > 0} -\beta(|p - b + \epsilon|) = -\beta(|p - b|)$. Therefore, $\lim_{b \uparrow p} V_B^0(b) = 0$. For $b > p$, the myopic optimal policy is $q = p$ from Theorem 4, and this yields reward $V_B^0(b) = 0$ for all $b > p$ from the third case of (44). This proves the continuity of $V_B^0(\cdot)$ at $b = p$ and completes the proof that $V_B^0(\cdot)$ is continuous on $[0, 1]$.

Applying Theorem 4.2 of Hinderer (2005), then yields the continuity of $V_B^*(\cdot)$. $\square$

Remark 4. Given the continuity of $V_B^*(\cdot)$ with respect to public belief and the fact that the control variable lies in the compact set $[0.5, 1]$, one might conclude that the optimal is guaranteed to exist at all public beliefs. However, there is an important subtlety which prevents such a conclusion: while $V_B^*(\cdot)$ is continuous with respect to public belief, it need not be continuous with respect to precision. Specifically, b_{i+1} is a function of both b_i and q_i (see Equation (3)); thus, $\mathbb{E}[V_B^*(b_{i+1})]$ may be denoted as $\phi(b_i, q_i)$. Now, b_{i+1} need not be continuous with respect to q_i , thus $\phi(\cdot)$ may not be continuous with respect to q_i . Thus, there may be regimes in which the optimal value is not attained by any policy. Indeed, we will show that such cases exist.

Proof of Theorem 5 We now leverage Lemma 16 to prove Theorem 5. We again let $z(b, q) = b + q - 2bq$ be the probability of an agent receiving signal B with precision q and prior public belief b and $y(b, q) = 1 - z(b, q)$ be the corresponding probability of the agent receiving signal G . We abuse notation to allow $V_B(b, q)$ to be the expected utility when beginning at public belief b , applying precision q in the current time step, and then following the optimal policy $\pi_B^*(\cdot)$ in all future steps. We can express $V_B(b, q)$ as follows by applying (13), (3), and (44):

$$V_B(b, q) = \begin{cases} -\beta(|q - p|) - Cz(b, q) \\ + \delta y(b, q) V_B^*(f(b, q, G)) \\ + \delta z(b, q) V_B^*(f(b, q, B)) & q \geq \max(b, 1 - b) \\ -\beta(|q - p|) \\ -C\mathbf{1}(b < 0.5) \\ + \delta V_B^*(b) & \text{otherwise} \end{cases} \quad (49)$$

Consider the latter case of (49) when $q < \max(b, 1 - b)$. Here, the only term affected by the precision policy is $-\beta(|q - p|)$, which is maximized by choosing q as close to p as possible. Thus, if the optimal precision is less than $\max(b, 1 - b)$, then it will be as close to p as possible.

We now consider four cases: (1) $b \in [0, 1 - p]$, (2) $b \in [1 - p, 0.5]$, (3) $b \in [0.5, p]$, and (4) $b \in (p, 1]$.

Case 1: $b \in [0, 1 - p]$, claims (A) and (B)

In this case, we will prove the existence of t_1 such that $\pi_B^*(b) = p$ for $b \leq t_1$. First note that if $\pi^*(b) < \max(b, 1 - b)$, we fall into the second case of Equation (49). Here, the only term affected by the precision policy is $-\beta(|q - p|)$, which is maximized by minimizing $|q - p|$. Because $p \in [0.5, 1]$, when $b \in [0, 1 - p]$, $p < \max(b, 1 - b)$. Thus, for $b \in [0, 1 - p]$, if $\pi_B^*(b) < \max(b, 1 - b)$, then $\pi_B^*(b) = p$.

Now, we evaluate the value function at belief $b = 0$. At $b = 0$, the state of the world is known to be B and, thus, no signal, regardless of its precision, will move the public belief. This can be seen mathematically by substituting $b = 0$ into the updated belief expressions in (13). Furthermore, at $b = 0$, if $q \geq \max(b, 1 - b)$, then $q = 1$. As argued above, if the precision is less than $\max(b, 1 - b)$, the optimal precision is $q = p$. We can compare the resultant value under these two candidate policies as follows from Equation (49):

$$V_B(0, 1) = -\beta(1 - p) - C + \delta V_B^*(0) \quad (50)$$

$$V_B(0, p) = -C + \delta V_B^*(0) \quad (51)$$

Because $V_B(0, 1) < V_B(0, p)$, $\pi_B^*(0) = p$ and $V_B^*(0) = V_B(0, p)$. Thus, from Equation (51), $V_B^*(0) = \frac{-C}{1-\delta}$. By Lemma 15, this is a lower bound for the value function at any public belief. Therefore, and since $V_B^*(b) \leq 0$, $|V_B^*(b)| \leq \frac{C}{1-\delta}$ for all $b \in [0, 1]$.

We now show that there exists an interval $[0, \epsilon)$ with $\epsilon > 0$ on which the optimum of the first case of Equation (49) is less than the optimum of the second case of Equation (49). Thus, in this interval the optimum control lies in the second case of Equation (49) so $\pi_B^*(b) < \max(b, 1-b)$. It follows from our initial argument earlier in this case that $\pi_B^*(b) = p$ in this interval, which would complete the proof.

We define function $g(\cdot)$ to be the maximum of the first case of the RHS of Equation (49):

$$g(b) = \max_{q \geq \max(b, 1-b)} \left[-\beta(|q-p|) - Cz(b, q) + \delta y(b, q) V_B^*(f(b, q, G)) + \delta z(b, q) V_B^*(f(b, q, B)) \right] \quad (52)$$

For $b \in (0, 1)$, $f(b, q, G)$ and $f(b, q, B)$ are both continuous with respect to q from Equation (13). Thus, the maximum in Equation (52) must exist because it is of a function which is continuous with respect to q over a compact set $q \in [\max(b, 1-b), 1]$.

We now take the limit of $g(\epsilon)$ as ϵ tends to 0. Because $q \in [\max(\epsilon, 1-\epsilon), 1]$, when $\epsilon \rightarrow 0$, $q \rightarrow 1$ (inside the maximum in Equation (52)). We now reason about each of the four terms in Equation (52). As $\epsilon \rightarrow 0$, $q \rightarrow 1$ so that 1) from the continuity of $\beta(\cdot)$, $\beta(|q-p|) \rightarrow \beta(1-p)$, 2) from the continuity of $z(\cdot)$, $z(\epsilon, q) \rightarrow 1$, $y(\epsilon, q) \rightarrow 0$. Thus, the second term goes to C . Because $V_B^*(\cdot)$ is bounded and $y(\epsilon, q) \rightarrow 0$, the third term goes to 0. Finally, because $z(\epsilon, q) \rightarrow 1$ and $f(\epsilon, q, B) \rightarrow 0$ and since $V_B^*(\cdot)$ is continuous from Lemma 16, the last term approaches $\delta V_B^*(0)$. Thus,

$$\lim_{\epsilon \rightarrow 0} g(\epsilon) = -\beta(1-p) - C + \delta V_B^*(0) \quad (53)$$

Applying precision p , however, yields the following from the second case of Equation (49):

$$\lim_{\epsilon \rightarrow 0} V_B(\epsilon, p) = \lim_{\epsilon \rightarrow 0} [-\beta(p-p) - C + \delta V_B^*(\epsilon)] \quad (54)$$

$$= -C + \delta V_B^*(0) \quad (55)$$

The last equality follows from the continuity of $V_B^*(\cdot)$ from Lemma 16.

Note that the expression in Equation (53) is strictly less than that in Equation (55). Thus, there exists an interval $[0, \epsilon)$ with $\epsilon > 0$ on which the optimum of the first case of Equation (49) is less than the optimum of the second case of Equation (49). The result follows.

Case 2: $b \in [1-p, 0.5)$, Claim (C)

We now argue that for $b_i \in [1-p, 0.5)$, $\pi^*(b_i) \geq 1-b_i$, yielding claim (C). Consider a stationary policy $\pi(\cdot)$ such that $\pi(b_i) < 1-b_i$. Note that, from Equation (3), the public belief will remain unchanged under such a policy. Thus, the same control will be applied at every future time step, and, from Equation (2), every agent will select action B . Using the facts that $1-b_i \leq p$ in this belief range, $\mathbb{P}(a_i = B | b_i, \pi(b_i)) = 1$, and $b_{i+1} = b_i$, we can apply Equation (6) to write the expected total discounted utility under such a policy as follows:

$$V_B^\pi(b_i) = -\beta(p - \pi(b_i)) - C + \delta V_B^\pi(b_i)$$

Thus, if this policy is applied, it will result in expected value $V_B^\pi(b_i) = \frac{-\beta(p - \pi(b_i)) - C}{1-\delta}$. That is, the cost of every agent choosing action B and continuing to apply precision $\pi(b_i)$. This is strictly lower than the value function lower bound of $\frac{-C}{1-\delta}$ established using Lemma 15 in Case 1 because $\pi(b_i) < 1-b_i \leq p$. Therefore, if an optimal policy exists at this point, it must be the case that $\pi_B^*(b_i) \geq 1-b_i$.

The remaining candidate policies are $q_i \in [1-b_i, 1]$. Applying Bellman's principle we can write the following:

$$V_B^*(b_i) = \sup_{q_i \in [1-b_i, 1]} r_B(b_i, q_i) + \delta \mathbb{E}[V_B^*(b_{i+1})] \quad (56)$$

Because $q_i \geq \max(b_i, 1-b_i)$ and $b \in [1-p, 0.5) \subset (0, 1)$, Equation (13) and Equation (3) imply that $b_{i+1} = \frac{b_i q_i}{y(b_i, q_i)}$ or $b_{i+1} = \frac{b_i(1-q_i)}{z(b_i, q_i)}$. Both of these expressions are continuous with respect to q_i and $V_B^*(\cdot)$ is continuous by Lemma 16, thus the RHS of Equation (56) inside the supremum is continuous with respect to q_i . Therefore, in Equation (56), the supremum over a compact set $q_i \in [1-b_i, 1]$ of a function which is continuous with respect to q_i must be attained, implying the existence of an optimal policy. Thus, $\pi_B^*(b_i) \geq 1-b_i$.

Case 3: $b \in [0.5, p]$ Claims (D) and (E)

For $b = 0.5$, $q \geq b$ as we assumed $q \in [0.5, 1]$. This places us in the first case of Equation (49). Furthermore, $\beta(\cdot)$, $y(b, q)$, $z(b, q)$, $f(b, q, G)$, and $f(b, q, B)$ are all continuous with respect to q for $b = 0.5$. Thus, the

supremum over $q \in [0.5, 1]$ of $V(b, q)$ must be attained as it is of a continuous function of q over a compact set of q . Therefore, an optimal policy must exist at $b = 0.5$.

Now consider $b \in (0.5, p]$ and a stationary policy $\pi'(\cdot)$ such that $\pi'(b) < b$. From Equation (3) and Equation (2), applying such a policy will result in public belief remaining unchanged, every agent selecting action G , and the same control being applied at every future time step. Using the facts that $b_i \leq p$, $\mathbb{P}(a_i = G | b_i, \pi'(b_i)) = 1$, and $b_{i+1} = b_i$, we can apply Equation (6) to write the total expected discounted utility under such a policy as follows:

$$V_B^{\pi'}(b_i) = -\beta(p - \pi'(b_i)) + \delta V_B^{\pi'}(b_i)$$

Thus, the policy results in expected value $V_B^{\pi'}(b_i) = \frac{-\beta(p - \pi'(b_i))}{1 - \delta}$. Note that, in this case, $V_B^{\pi'}(b_i)$ is an increasing function of $\pi'(b_i)$. Thus, the supremum of $V^*(\pi')_B(b_i)$ over $\pi'(b_i) \in (0.5, b)$ is not attained.

We now consider two cases: (1) $\max_{q_i \in [b_i, 1]} V(b_i, q_i) \geq \sup_{\pi'(b_i) \in (0.5, b_i)} \frac{-\beta(p - \pi'(b_i))}{1 - \delta}$, and (2) the complement. In the former case, we can argue as we did for $b = 0.5$ that the supremum must be attained and an optimal policy exists such that $\pi_B^*(b_i) \geq b_i$. In the latter case, as argued above, no optimal policy exists. Instead, for sufficiently small $\epsilon > 0$, there exists an ϵ -optimal policy such that $\pi_B^\epsilon(b_i) = b_i - \epsilon$. This yields claims (D) and (E).

What remains is to show that there exists $t_2 \in (0.5, p)$ such that for $b_i \in (t_2, p]$ we fall into the latter case, yielding claim (F). We define functions $m(\cdot)$ and $n(\cdot)$ as follows corresponding to the resultant value from choosing precisions less than b or greater than or equal to b , respectively:

$$\begin{aligned} m(b) &= \sup_{\pi'(b) \in [0.5, b)} \frac{-\beta(p - \pi'(b))}{1 - \delta} \\ n(b) &= \max_{q \geq b} V(b, q) \end{aligned}$$

Taking the limit of $m(\cdot)$ as $b \rightarrow p$ yields:

$$\lim_{b \rightarrow p} m(b) = 0 \tag{57}$$

When $q \geq b$ we fall into the first case of Equation (49). Thus we can expand $n(b)$ as follows:

$$\begin{aligned} n(b) &= \max_{q \geq b} [-\beta(|q - p|) - Cz(b, q) \\ &\quad + \delta y(b, q)V_B^*(b, q) \\ &\quad + \delta z(b, q)V_B^*(b, q)] \end{aligned} \tag{58}$$

We will argue that, for $b \in (0.5, p]$ and $q \geq b$, $n(b) \leq -C(1 - p)$. Since p is a fixed parameter strictly less than 1, this means from Equation (57) that $n(b) \leq -C(1 - p) < \lim_{b \rightarrow p} m(b)$, which implies the existence of an open interval $(t_2, p]$ where an optimal policy does not exist, yielding claim (F). We now argue that, for $b \in (0.5, p]$ and $q \geq b$, $n(b) \leq -C(1 - p)$. Note that every term of Equation (58) is bounded above by 0, so it suffices to show that any one term, say the second term $-Cz(b, q)$, is upper bounded by $-C(1 - p)$. First note that $z(b, q) = q(1 - 2b) + b$ so $z(b, q)$ is decreasing in q since $b > 0.5$. Thus, $z(b, q) \geq z(b, 1) = 1 - b \geq 1 - p$. The last inequality follows because $b \in (0.5, p]$. The claim follows since $p < 1$.

Along with Equation (57), this implies the existence of an open interval (t_2, p) where an optimal policy does not exist, yielding claim (F).

Case 4: $b \in (p, 1]$ Claim (A)

For $b \in (p, 1]$, consider applying stationary policy $\pi(\cdot)$ such that $\pi(b) = p$. Because $b > p$, Equation (3) and Equation (2) imply that such a policy results in public belief remaining unchanged, every agent selecting action G , and the same control being applied at every future time step. We can apply Equation (6) to write

$$V_B^\pi(b) = -\beta(p - p) + \delta V_B^\pi(b) = \delta V_B^\pi(b)$$

Thus, $V_B^\pi(b) = 0$. Since this is also an upper bound of the optimal value function for any b , $\pi_B^*(b) = p$. ■

C.9 SOCIAL WELFARE MONOTONICITY

Lemma 17. *Precision Monotonicity of Social Welfare*

If $\pi_1(b) \leq \pi_2(b) \forall b \in [0, 1]$, for policies $\pi_1(\cdot), \pi_2(\cdot)$, then

$$W^{\pi_1}(b) \leq W^{\pi_2}(b) \forall b \in [0, 1]$$

Proof. The one-step welfare term $-C \min(b_i, 1 - b_i, \pi(b_i))$ is increasing in $\pi(b_i)$.

Moreover, convexity of social welfare with respect to public belief follows from the same argument used in the proof of Theorem 2, because social welfare coincides with the altruistic value function in the special case $\beta(\cdot) = 0$.

The proof then follows by applying the same reasoning as in Lemma 9 (Appendix C.4), which shows that increasing precision increases welfare. $\square$

D EXTENSION TO HETEROGENEOUS AGENT PREFERENCES

We argue here that, for the altruistic planner, the baseline problem is equivalent to a formulation with two observable types of individuals who have opposing preferences.

Assume that there are two agent types, G and B , and denote agent i 's type by t_i . The first type receives unit reward when $a_i = \omega$ and no reward otherwise, whereas the second receives unit reward when $a_i \neq \omega$ and no reward otherwise. Further suppose that every agent observes her own type and the types of all other agents. All remaining aspects of the model are unchanged.

Agent i 's private-belief update is given below. It remains unchanged because it depends only on the signal structure, namely $\mathbb{P}(s_i = \omega) = q_i$.

$$\tilde{b}_i = \mathbb{P}(\omega = G | \mathcal{H}_i, q_i, s_i) = \begin{cases} \frac{q}{1+2b_iq_i-b_i-q_i} b_i & s_i = G \\ \frac{1-q}{b_i+q_i-2b_iq_i} b_i & s_i = B \end{cases} \quad (59)$$

Using Equation (59), agent i 's action can be written as follows, with the superscript indicating agent i 's type and $!$ denoting negation:

$$a_i^G = \begin{cases} s_i & 1 - q_i \leq b_i \leq q_i \\ G & q_i < b_i \\ B & b_i < 1 - q_i \end{cases}$$

$$a_i^B = \begin{cases} !s_i & 1 - q_i \leq b_i \leq q_i \\ B & q_i < b_i \\ G & b_i < 1 - q_i \end{cases}$$

Because each agent's type is publicly known, all agents can perfectly infer agent i 's private signal whenever $q_i \geq \max(b_i, 1 - b_i)$. If $q_i < \max(b_i, 1 - b_i)$, agent i 's action is uninformative.

Consequently, after observing agent i 's type t_i , precision q_i , and action a_i , the public belief is updated as follows:

$$b_{i+1} = f(b_i, q_i) = \begin{cases} \tilde{b}_i & 1 - q_i \leq b_i \leq q_i \\ b_i & \text{o.w.} \end{cases} \quad (60)$$

In this setting, an altruistic planner seeks to induce type- G agents to choose ω and type- B agents to choose $!\omega$, because those choices align with the agents' respective utility functions.

The planner's instantaneous reward can therefore be expressed as follows, with the superscript denoting agent i 's type:

$$r^G(b_i, q_i) = -\beta(|q_i - p|) - C \mathbb{P}(a_i \neq \omega)$$

$$= \begin{cases} -\beta(|q_i - p|) - C(1 - q_i) & q_i \geq \max(b_i, 1 - b_i) \\ -\beta(|q_i - p|) - C \min(b_i, 1 - b_i) & q_i < \max(b_i, 1 - b_i) \end{cases}$$

$$r^B(b_i, q_i) = -\beta(|q_i - p|) - C \mathbb{P}(a_i = \omega)$$

$$= \begin{cases} -\beta(|q_i - p|) - C(1 - q_i) & q_i \geq \max(b_i, 1 - b_i) \\ -\beta(|q_i - p|) - C \min(b_i, 1 - b_i) & q_i < \max(b_i, 1 - b_i) \end{cases}$$

These expressions are identical to each other and equal to the reward function in Equation (5), establishing equivalence of the altruistic planner's problem under heterogeneous but observable types. $\blacksquare$

E EXPERIMENTS

E.1 EXPERIMENTAL SETUP

To simulate controlled social learning, we consider a sequence of agents deciding whether to buy a new car model, with decisions informed by observations of past actions and by an advertisement whose precision is selected by the planner. LLMs serve three distinct roles: (1) agents, (2) planner, and (3) oracle. Each agent receives a profile containing demographic information, anticipated car-use cases, and preferences. At the outset, we designate either the good or bad car as the offering. The information sets and objectives of the planner and agents match those specified in Section 3.

The oracle has two functions: (1) it generates a generic message that serves as the baseline-precision signal absent planner intervention, and (2) after the planner chooses a precision, it generates a target-agent-specific message with the prescribed precision.

The oracle receives a fact sheet for each car type, including attributes such as safety rating and customer satisfaction, together with the agents' profiles. The fact sheet and profile jointly support generation of a tailored message at the precision prescribed by the planner. Additional details on setup and prompting appear in Appendix E.2.

Each agent makes a binary decision using observations of previous decisions and her private signal, with the objective of matching the true parameter value.

The planner receives the same decision history and chooses a precision level for each agent's message. It is also given the precision of a generic, untailored message that can be sent at zero cost. In the altruistic case, the planner's objective is to induce correct decisions; in the biased case, the planner's objective is to induce agents to buy the car.

We compare the LLM simulations against analytical solutions computed via JuliaPOMDP (Egorov et al., 2017).

E.2 PARAMETERS AND PROMPTS

E.2.1 INSTANCE PARAMETERS

Experiments use the LG AI EXAONE Deep 32B model (LG AI Research, 2025). We solve the model analytically and run LLM simulations for the parameter values reported in Table 1.

Parameter	Description	Values
C	Cost of a bad action	1
k	Scaling factor for control cost	{0.1, 0.3, 0.5, 0.7, 0.9}
p	Baseline signal precision	{0.6, 0.7, 0.8, 0.9}
δ	Discount Factor	{0, 0.25, 0.5, 0.75, 1}

Table 1: Parameter values used in the analytical and LLM-simulation instances.

The car facts are constructed so that the good car is desirable for every agent and the bad car is undesirable for every agent. The GOOD and BAD fact sheets are reported below, respectively.

```
fact_sheet["GOOD"] = [
    "EPA-rated range: 290 miles.",
    "Fast-charge 10-80 %% in 24 min on 150 kW DC.",
    "8-year / 100,000-mile battery warranty.",
    "NHTSA 5-Star overall safety rating.",
    "Cold-soak -5 °F keeps 93 %% range.",
    "Cools cabin 100→72 °F in 3 min.",
    "Cargo volume 26 ft3 total.",
    "Tow's 1 500 lb with factory hitch.",
    "Average 1.2 service visits first 3 yrs.",
    "92 %% owners would buy again (Apr-25).",
    "Qualifies for federal credit.",
    "Quarterly OTA updates incl. battery pre-heat."
]
fact_sheet["BAD"] = [
    "EPA-rated range: 155 mi (combined).",
    "Fast-charge 10 %% → 80 %% in 51 min (50 kW DC hardware limit).",
```

```

1836         "Battery warranty: 3 yr / 36 000 mi, first owner only.",
1837         "NHTSA overall safety rating 3 Stars (frontal-offset weakness).",
1838         "Cold-soak 15 °F retains 58 %% range;cabin heat cuts range 20%.",
1839         "Cabin cools 105 °F → 72 °F in 11 min.",
1840         "Cargo volume 17 ft3; rear seats fixed (no fold-flat).",
1841         "No factory towing approval.",
1842         "Average 3.8 service visits per year (battery-alert issues).",
1843         'Owner survey: 41 %% "would buy again" (1 200 owners, Feb-25).',
1844         "Not eligible for federal tax credit.",
1845         "OTA updates limited to infotainment."
1846     ]

```

Profiles were generated programmatically. Profiles included an agent's name, age, city of residence, interests, planned use case for the car (e.g., commuting, road trips, etc.), and major pain points or concerns (e.g., fuel efficiency, winter driving, etc.).

E.2.2 PROMPTS

Agent LLM Prompt:

```

1853 sys = (
1854     f"You are deciding whether to buy a car. "
1855     f"Your profile is {profile}"
1856     f"You start believing there's a 50% chance the car is GOOD. "
1857     f"You have two information sources: "
1858     f"1. Previous agents' decisions "
1859     f"2. An advisor's message about the car "
1860     f"Your goal is to maximize your reward "
1861     f"i.e. BUY if the car is GOOD and "
1862     f"NOT BUY if the car is BAD. "
1863     f"If you BUY a GOOD car, you GAIN ${C}. "
1864     f"If you NOT BUY a BAD car, you GAIN ${C}. "
1865     f"If you BUY a BAD car, you LOSE ${C}. "
1866     f"If you NOT BUY a GOOD car, you LOSE ${C}. "
1867     f"belief = your probability (0.0-1.0) that the car is GOOD. "
1868     f"decision = exactly 'BUY' or 'NOT BUY'."
1869 )
1870 user = (
1871     f"PREVIOUS DECISIONS:\n{history_txt or '(none)'}\n\n"
1872     f"ADVISOR MESSAGE:\n{message}\n\n"
1873     f"TASK: Use the previous decisions and the advisor message. "
1874     f"Decide BUY or NOT BUY. "
1875     f"Update your belief about car quality (0.0-1.0). "\n"
1876     f'JSON response:{{"belief": <number>,"decision": "<BUY or NOT BUY>"}}'
1877 )

```

Planner LLM Prompt:

```

1878 sys = (
1879     f"You are providing messages to potential car buyers. "
1880     f"Each potential car buyer (or agent) has the same information. "
1881     f"Your goal is to get agents to {objective}."
1882     f"Each decision according to your objective, earns you {C} dollars. "
1883     f"You and the agents have the same information. "
1884     f"Initially there is a 50% chance the car is GOOD. "
1885     f"The agents' aim to BUY if the car is GOOD "
1886     f"and NOT BUY if the car is BAD. "
1887     f"You can send a baseline message for free "
1888     f"or tailor the message to the agent at a cost. "
1889     f"Tailoring costs {k} dollars times the change in precision. "
1890     f"(Message precision is the probability (0.5 to 1.0) "
1891     f"that the message leaves the buyer with the correct impression. "
1892     f"For example, if the car is likely GOOD: "
1893     f"a high precision message will likely suggest BUY "
1894     f"and a low precision message will likely suggest NOT BUY. "

```

```

1890     f"You may choose to increase or decrease precision. "
1891     f"Agents decide based on your message and the decision history. "
1892     f"They update beliefs by observing previous decisions. "
1893     f"IMPORTANT: Thus your objective is to maximize:"
1894     f"{C} * (# of agents who choose according to your objective) "
1895     f"- sum of ({d}^i*{k}*abs(message_precision - baseline_precision))\n"
1896     f"where {d} is your discount factor for future cost. "
1897 )
1898 user = (
1899     f"BASELINE MESSAGE precision:\n{pi_base:.2f}\n"
1900     f"DECISION HISTORY:\n{history_txt}\n"
1901     f"TASK:"
1902     f"Based on the history and your objective, "
1903     f"choose your desired precision.\n"
1904     f'JSON response: {"message_precision": <number>}\n'
1905 )

```

Oracle LLM Prompt:

```

1906 sys = (
1907     f"You will be given "
1908     f"1. an agent profile, "
1909     f"2. information about a car, "
1910     f"3. the true quality of the car, "
1911     f"4. a baseline message and the 'precision' of that message, and "
1912     f"5. a desired precision for the message you output. "
1913     f"(Message precision is the probability (0.5 to 1.0) "
1914     f"that the message leaves the buyer with the correct impression. "
1915     f"For example, if the car is likely GOOD: "
1916     f"a high precision message will likely suggest BUY "
1917     f"and a low precision message will likely suggest NOT BUY. "
1918     f"IMPORTANT: Output a message with the requested precision. "
1919     f"IMPORTANT: The message should be tailored to the profile. "
1920 )
1921 user = (
1922     f"AGENT PROFILE:{profile}\n"
1923     f"TRUE CAR QUALITY: {omega}\n\n"
1924     f"CAR FACTS: {car_facts}\n"
1925     f"BASELINE MESSAGE:{baseline_message}\n"
1926     f"BASELINE MESSAGE precision: {p}\n"
1927     f"DESIRED MESSAGE precision: {q}\n"
1928     f"IMPORTANT: Output a message with the requested precision. "
1929     f"TASK: Write a message (MAX 100 words) with the desired precision."
1930     f'JSON response: {"message": "<message>"}\n'
1931 )

```

1929 E.3 VALIDATION OF LLM BELIEFS AND ORACLE

1930 We now present additional experiments validating the LLMs' self-reported beliefs, as shown in figure 1b, and
1931 the performance of the LLM oracle. In these experiments, we treat LLMs as rational and utility-maximizing,
1932 though not necessarily Bayesian, agents. The experiments have two aims: (1) to show that the model's self-
1933 reported posterior corresponds to the belief governing its choices in state-contingent decision problems, and (2)
1934 to verify that oracle-generated messages possess the requested precision.

1935 First, we provide the LLM with a prior and a new signal, elicit its self-reported posterior belief, and then ask
1936 it to make a binary "buy/not-buy" decision that yields unit reward if and only if the true state is good. A
1937 rational agent should choose "buy" exactly when its subjective belief exceeds 0.5. As shown in figure 4a, the
1938 LLM's decisions align sharply with this threshold: it overwhelmingly chooses "buy" when its reported posterior
1939 exceeds 0.5 and "not buy" otherwise. Next, we elicit the model's willingness to pay for a payoff contingent on
1940 the true state. Across a range of prices c , we ask whether the agent would pay c in exchange for unit utility
1941 if the state is good. A rational agent with belief p should accept exactly when $c < p$. Figure 4b plots the
1942 fraction of "buy" decisions across posterior-price pairs and shows a clear separation: acceptance rates are high
1943 when $c < p$ and low when $c > p$, precisely as expected from a utility-maximizing decision maker with belief
1944 p . Together, the two experiments indicate that the self-reported posterior closely tracks the model's operative
1945 decision-making belief.

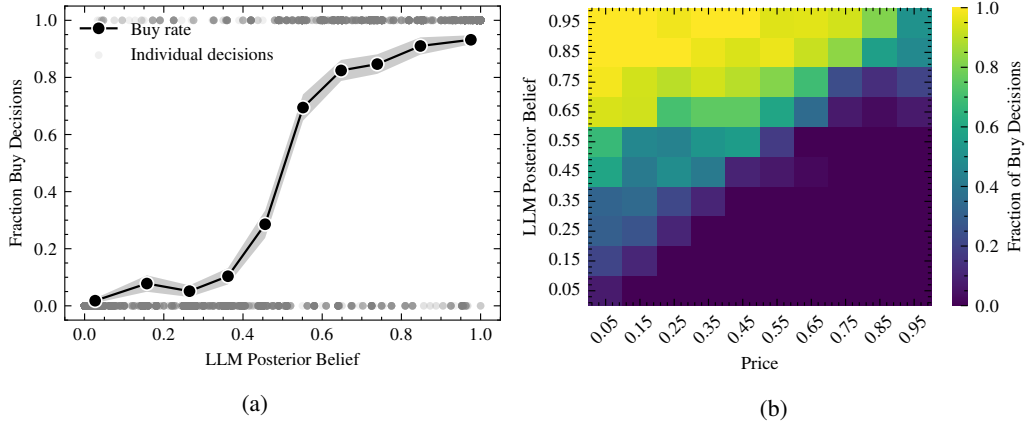


Figure 4: LLM ‘buy’ rates as functions of (a) self-reported posterior belief and (b) self-reported posterior belief together with the purchase price.

We similarly validate the oracle’s ability to craft messages at a specified precision. After the oracle generates a message with target precision q , we ask another LLM to estimate the message precision $\hat{q}$. We conduct this procedure for $q \in \{0.5, 0.6, 0.7, 0.8, 0.9, 1.0\}$ across 100 programmatically generated agent profiles, with 10 trials per combination, for a total of 6,000 trials. The mean absolute error $|q - \hat{q}|$ is only 0.025, indicating that realized oracle-message precision closely follows the requested precision.

E.4 IMPACT OF NON-BAYESIAN LLM BELIEF UPDATES ON LEARNING TRAJECTORIES

We now examine belief trajectories from a representative LLM experiment to clarify the effect of non-Bayesian belief updating. In this exercise, we fix the policy to the analytically optimal policy and apply it to LLM-simulated agents. When the car is bad (right panel), the altruistic planner drives the Bayesian belief (dotted black line) rapidly toward 0. The LLM belief (solid black line), however, is more persistent. This aggregate trajectory follows directly from the micro-level biases: public belief resists entering a cascade (NB3) because individual agents overreact to occasional positive signals that run counter to the group’s increasingly negative belief (NB2).

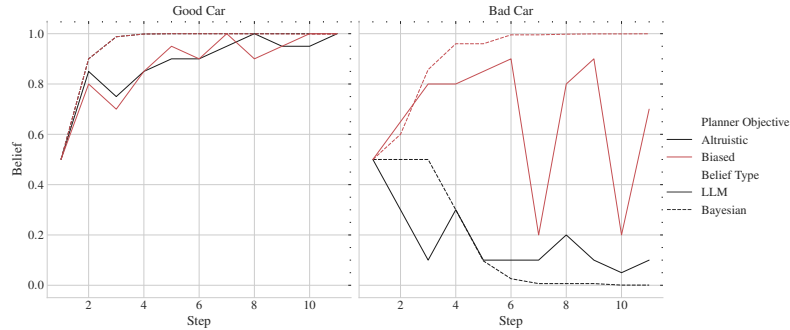


Figure 5: Agent belief-updating trajectories for the true Bayesian belief and the LLM-reported belief under analytically optimal policies, shown across planner objectives and true states of the world.